A total-variation floor and a rank-three classification for the parity-compressed sector of a log-singular Toeplitz form with a prime-power comb

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Abstract

This paper is about the near-degeneracy behind a Toeplitz spectral floor: a total-variation bound, three linear functionals, and a priced table of what the standard unconditional toolbox reaches. The object is the reflection-odd compression A of a finite Toeplitz form whose lag sequence is a logarithmic profile minus an explicit comb, and the question is how small the Rayleigh quotient of A can be on the low end of the parity spectrum. Four results are proved. First, a *total-variation floor*: every trial vector whose leading parity coordinate is normalised to one has lag weights of total variation at least $1/\mu_1$, where $\mu_1 = 4\sin^2(\pi/(2h+1))$ is the spectral gap of the parity compression of the second difference; the floor is closed, involves no comb node, no taper and no truncation, and it prices the entire trial-vector route at $O(h^{-2})$ for every vector of bounded value. Second, a *gauge rigidity* statement: the leading-coordinate normalisation pins only the scale of the trial direction, both the value and the total variation are homogeneous of degree two, hence their ratio is a function of the ray and takes the same range in every reparametrisation of the entry spectrum — no sector change removes the floor, and a shifted sector conserves it exactly, floor times transfer being $1/\mu_1$ identically. Third, an exact *rank-three classification*: the two-by-two raw Gram block of the low parity modes splits as an archimedean block minus a comb block, the determinant polarises into three terms, and the comb–comb term is the polarisation of the determinant on symmetric two-by-two matrices — a form of rank three and signature $(1, 2)$ — so the comb enters the determinant only through *three* linear functionals of its weights, for every window and every truncation. Fourth, a classification of what this costs: any majorant of the comb–comb term built from a norm of the weight vector must dominate the supremum of a rank-three quadratic over the image of that norm ball, which is where the priced routes lose. The remaining question — whether the two rows of the block become collinear at a rate uniform in the window index — is stated as an open problem about a finite bilinear form. Around it we report, as measurements on declared surfaces: the rates; a table of five unconditional majorants together with the exponent each certifies, the best of them short of the demanded one by two units in the exponent; five structural obstructions, each exhibited failing; a curve of the binned exponent against comb density over four decades of density, whose densest bins are consistent with zero and whose undecidability under the declared caps is part of the claim; and two controlled placement experiments, the second of which closes into an exact identity for the response of the ratio to a displacement of comb nodes inside their lag cells. Randomising the node positions at fixed weights leaves every theorem here intact and removes the phenomenon: the arithmetic enters the theorems only as the source of the comb positions and weights, and every statement proved here holds for an arbitrary finite comb.

1 Introduction

1.1 The problem, and why the direction is hard

Let $c = (c_m)_{m=0}^{M-1}$ be a real lag sequence, let $T_M(c)$ be its symmetric Toeplitz matrix, and let A be the compression of $T_M(c)$ to the reflection-odd sector of the window (Section 2). The question this paper is about is a lower bound for the Rayleigh quotient of A on the low end of a fixed orthonormal mode family — equivalently, an upper bound for the extent to which the quadratic form $v \mapsto v^{\mathsf{T}} A v$ can cancel along a smooth direction.

The classical route from a Toeplitz form to a spectral bound is the symbol. If $\ell \le \sigma$ pointwise then $T_M(\ell) \preceq T_M(\sigma)$ by Parseval, and a positive pointwise minorant of the symbol transfers to a positive floor for the form [23, 10, 15, 6, 7]. For the class considered here that route is unavailable, and not marginally so: the symbol is a logarithmic profile minus an explicit cosine comb, the comb dips are numerous and narrow, and on every window of the declared surface of Section 8 the infimum of the symbol is negative, while the form restricted to the parity sector is positive. The smallest Rayleigh quotient is therefore not produced by a small value of the symbol at some frequency; it is produced by *cancellation inside the quadratic form*, along a direction that lives in the low modes. A pointwise minorant is by construction blind to which vector realises the minimum.

The natural substitute is a trial-vector route: exhibit an explicit direction, evaluate the form on it, and price the evaluation. This paper is a classification of that route. Its first two results say that the route has a closed price which no reparametrisation removes; its second two say that the object which remains after the price is paid is a quadratic polynomial in three linear functionals of the comb weights, and that this is exactly why the unconditional toolbox — size budgets, symbol arguments, sectors and gauges, splits, perturbation theory, the bilinear sieve box — does not reach it. What is *not* claimed anywhere is a bound on the arithmetic: the three functionals are finite sums over the comb, all four results hold for an arbitrary finite comb, and the one remaining question is stated as an open problem in Section 7.

The companion paper [11] treats the same window form from the opposite direction — a certified lower bound for a Galerkin defect on a nested ladder — and we cite it for the window form and the comb class rather than repeating them.

1.2 Contribution

(C1) **The total-variation floor, and the price it fixes** (Theorem 3.1 with Corollaries 3.2–3.4). For every trial vector whose leading parity coordinate is normalised to one, the lag weights w of the associated window vector satisfy $\mathrm{TV}(w) \ge |w_0| = \|v\|^2 \ge 1/\mu_1 \ge N^2/(4\pi^2)$, where $\mu_1 = 4\sin^2(\pi/N)$, $N = 2h+1$, is the spectral gap of the parity compression L_P of the second difference [15]. The proof is four steps and uses no property of the comb whatsoever. The load-bearing consequence is a *price*: the ratio of the value to the total variation is at most μ_1 times the value, so every trial vector of bounded value certifies a ratio $O(h^{-2})$, and the boundary-free summation by parts [1] can certify nothing better.

(C2) **Gauge rigidity of the normalisation** (Theorem 4.1 with Propositions 4.3–4.4 and Corollary 4.2). The normalisation pins only the scale of the trial direction; value and total variation are homogeneous of degree two; hence their ratio is a function of the ray, and its range over admissible vectors is the *same* for every entry spectrum. A shifted spectrum does flatten the floor, but the transfer factor that returns the bound to the original scale conserves the product exactly: floor times transfer is $1/\mu_1$ at every shift. So (C1) cannot be reparametrised away, and the parity compression is not a convenience — Proposition 4.4 shows it is what makes the normalisation well posed at all.

(C3) **The exact bilinear form and the rank-three classification** (Section 6: Propositions 6.1 and 6.2, Theorem 6.3, Corollary 6.6). The two-by-two raw Gram block of the two lowest parity modes splits exactly as $\widehat{A}_2 = B - S$ with B archimedean and S a weighted sum of comb reads; the determinant polarises as $\det \widehat{A}_2 = \det B - D(B, S) + \det S$; and D is the polarisation of det on symmetric two-by-two matrices, a form of rank three and signature $(1, 2)$ in the coordinates (X_{11}, X_{22}, X_{12}). Consequently the comb–comb term is $S_{11}S_{22} - S_{12}^2$: a quadratic polynomial in *three* linear functionals of the comb weights, for every window, every h and every truncation. Corollary 6.6 turns this into the statement about majorants that the priced routes of Section 7 run into.

(C4) **A priced table, a density curve, and one open problem** (Section 7). What remains after (C1)–(C3) is a single question about a finite bilinear form: is the near-collinearity of the two rows of $\widehat{A}_2$ bounded uniformly in the window index? It is stated as Problem 7.1 in three equivalent forms — never as an approach to, a weakening of, or a consequence of any conjecture. Around it, Proposition 7.5 proves five unconditional majorants and Measurement 7.6 reports the exponent each certifies on a declared surface; Measurement 7.7 exhibits five further routes failing for structural reasons; Measurement 7.9 gives the binned exponent of Definition 7.8 against comb density over four decades, with Remark 7.10 stating what that curve cannot decide and at what price; and Measurements 7.11 and 7.14 locate the phenomenon by intervention, the second through the closed-form response of Proposition 7.13.

Two of these four are theorems; two are classifications supported by theorems together with declared measurements. Which is which is recorded in Section 1.3 rather than left to the reader.

1.3 Typing of the results

Every numbered statement of this paper that makes a claim carries exactly one of three types, and the type is part of the statement. Table 1 types every one of them: a definition and the open problem are marked as what they are, and a remark that only reads results already stated carries a dash, since it claims nothing of its own.

proved

A written proof is given (or, for classical facts, cited). The statement holds for every window and every finite comb within its stated hypotheses. Where a hypothesis is itself of a different type — positive definiteness of A on the parity sector, for instance — the statement is *conditional* and says so.

certified per window

The statement is verified on each window of a declared finite surface by a computation whose failure mode is loud (a completed Cholesky factorisation, an exact identity residual against a declared floor). No quantifier over windows is claimed.

measured

The statement is a number, a fitted exponent, or a controlled experiment on a declared finite surface, reported as such. Fitted exponents are never used as bounds.

Table 1: Typing of every numbered statement. "Conditional" marks a proved implication whose hypothesis is of a weaker type; the hypothesis is then named in the statement.

statement	type	remark
Section 2 — setting and scaffolding		

continued on the next page

statement	type	remark
the comb is not used (Rem. 2.1)	—	the scope statement: Secs. 2–6 hold for an arbitrary finite comb
parity spectrum (Lem. 2.2)	proved	closed eigenpairs and the gap (6); Kac–Murdock–Szegő [15]
L1 interlacing (Lem. 2.3)	proved (cited)	Cauchy [3], Courant–Fischer [5]
L2 Schur nesting (Lem. 2.4)	proved	Schur [21], Haynsworth [12]
L3 constrained minimum (Lem. 2.5)	proved	requires $G_k \succ 0$
gauge law (Lem. 2.9)	proved	$\sum_d w_d = 0$, unconditional
Abel swap (Lem. 2.10)	proved	boundary-free, $w_M = 0$
C5 cascade forms (Prop. 2.6)	proved	prefix / minor / regression
C6 diagonal invariance (Lem. 2.7)	proved	one line
B4 cascade direction (Prop. 2.8)	*conditional*	proved given $\widehat{A} \succ 0$; the hypothesis is certified per window
Section 3 — the total-variation floor		
A the floor (Thm. 3.1)	proved	comb-free, taper-free
A1 the price (Cor. 3.2)	proved	the load-bearing corollary
A2 Abel/Hölder (Cor. 3.3)	proved	
A-bis (Cor. 3.4)	proved	no normalisation needed
A4 negative control (Rem. 3.5)	proved	one line
A3 sharpness (Meas. 3.6)	measured	μ_1 TV $\in [5.00, 23.20]$ on Surface II
Section 4 — gauge rigidity		
B homogeneity (Thm. 4.1)	proved	
B1 floor × transfer (Prop. 4.3)	proved	pure shift
B2 the null mode (Prop. 4.4)	proved	closed spectra
no escape (Cor. 4.2)	proved	
sector battery (Meas. 4.5)	certified per window	$35 \times 6 \times 27$, residual $\leq 8.8 \cdot 10^{-10}$
Section 5 — the two-dimensional reduction		
C1 exact $K = 2$ (Prop. 5.1)	proved	
C2 pivot identity (Prop. 5.2)	proved	
C3 union identity (Prop. 5.4)	proved	
C7 Wronskian (Lem. 5.6)	proved	the corner entry 3 closes it
C8 maximal minors (Lem. 5.7)	proved	Lagrange [17]
C9 determinant reading (Lem. 5.8)	proved	uses no positivity
C10 Gershgorin (Lem. 5.9)	proved	uniform in the window
C11 det-collapse (Prop. 5.10)	proved / *conditional*	the closed form is unconditional; the harmonic-mean bound needs $\widehat{A}_2 \succ 0$
wrong sign priced (Rem. 5.3)	proved	the identity (30), $\rho = 4r_{12}^2/(1 - r_{12}^2)$
one obstruction, three readings (Rem. 5.5)	proved	the explicit factor $1 + 3r_{12}^2 \in [1, 4]$; the reading is an observation

continued on the next page

statement	type	remark
the section read together (Rem. 5.11)	—	which statements are identities and which two are not
Section 6 — bilinear form and rank		
D0 exact split (Prop. 6.1)	proved	needs an admissible resolution
D1 polarisation (Prop. 6.2)	proved	
D rank three (Thm. 6.3)	proved	every h, every truncation
D wedge form (Lem. 6.4)	*conditional*	proved given rank-one reads; the defect is measured
D3 majorants (Cor. 6.6)	proved	the class $\mathfrak{M}$ of Def. 6.5; the route reading of Rem. 6.7 is a reading
D3′ ℓ^1 mass (Meas. 6.8)	measured	40.6× the signed value on the declared subsample; the rank-one defect priced on both surfaces
D4a closed weights (Lem. 6.9)	proved	one formula on $1 \le d \le M-1$
D4b bandwidth K (Lem. 6.10)	proved	the $4K$-dimensional band space (52)
D4c band ⇒ η-rank (Lem. 6.11)	proved	general; $r = (2\Omega+1)(q+1)$ for Ω frequencies and degree bound q
D4d spline read (Lem. 6.12)	proved	$\eta = O(K^2/N^2)$
D4 Type II η-rank (Prop. 6.13)	proved	$\sigma_{4K+1} \le \eta\sqrt{RC}$; needs the interior hypothesis (57)
D4″ what the interior hypothesis costs (Rem. 6.14)	—	without (57) the conclusion is false as stated; the triangular cut-off is not of low rank
D4′ Type II collapse (Meas. 6.15)	measured	the observed 2 is sharper than the proved 8 and stays measured
D5 reference window (Meas. 6.16)	measured	the three-term cancellation
Section 7 — the open problem		
E0 Problem 7.1	*open*	three equivalent forms; a question about a finite bilinear form
E1 angle against determinant (Lem. 7.2)	proved	the factor lies in $(0, 1]$; used only to relate two readings
E2 rates (Meas. 7.3)	measured	fitted exponents; every surface and both readings declared
E3a dual-point read (Lem. 7.4)	proved	the two points $\gamma_k = 2\pi k/(ND)$ and their spacing are derived, not assumed

continued on the next page

statement	type	remark
E3 five majorants (Prop. 7.5)	proved	five unconditional inequalities with their hypotheses
E4 priced routes (Meas. 7.6)	measured	the exponent each proved route certifies on Surface II
E5 structural obstructions (Meas. 7.7)	certified per window	each route exhibited failing on Surface I
E6 the binned exponent (Def. 7.8)	definition	deterministic; ladder-dependent by Rem. 7.10
E7 density curve (Meas. 7.9)	measured	on Surface III, quoted with its rung ladder
E8 undecidability, fence, caveat (Rem. 7.10)	measured	part of the claim of E7, not a caveat on it
E9 the scramble (Meas. 7.11)	measured	controlled experiment at fixed weight multiset
E10 exact placement derivative (Lem. 7.12)	proved	$\dot{c}$ affine and piecewise constant
E11 placement response (Prop. 7.13)	proved	Hellmann–Feynman [13, 8]; simple extreme eigenvalues
E12 causality and the pole (Meas. 7.14)	measured	a derivative at a point on named instances, not a rate
E13 what is not claimed (Rem. 7.15)	—	the binding negative list
Section 8 — numerical verification		
F1–F3 conventions (Sec. 8.1)	—	floors, conditioning cap, mutation controls
the three surfaces (Sec. 8.2)	—	declared before any number is read
cross-reference (Tab. 3)	—	every numbered statement to its check
the two fit remarks (Rem. 8.1, 8.2)	measured	why $h^{+1.997}$ is the closed h^2; why two exponents are one object

Table 1 is the first thing to read. Four entries deserve emphasis. Proposition 2.8 and the harmonic-mean half of Proposition 5.10 are *conditional*: they are proved implications whose hypothesis is positive definiteness of the raw Gram block, which on the surfaces of Section 8 is a numerical fact about finitely many finite matrices and nothing more. Lemma 6.4 is conditional in the same sense, on a rank-one hypothesis whose defect is measured. And Measurement 6.15 is the one place where a phrase of the source material does not convert: "effective rank $O(1)$" is not a theorem. What is proved instead is the chain Lemmata 6.9–6.12 and Proposition 6.13, which bounds the $(4K+1)$-st singular value of a factorisation-indexed block by a declared truncation error under a declared interior hypothesis; the measured number of significant singular values is smaller than what that proposition gives, and it therefore stays measured.

Two entries in the last block of the table need the same care. Measurement 7.6 carries two types in one row on purpose: the inequality of each route is proved (Proposition 7.5), and only the exponent it certifies is a fit on Surface II. And Measurement 7.9 is quoted with its rung ladder, because the estimator of Definition 7.8 is ladder-dependent: a bin value reported without the ladder it was computed on is not reproducible (Remark 7.10).

symbol	meaning
$D,\ M \in 2\mathbb{N},\ \alpha = MD/2,\ h = M/2,\ N = 2h+1$	cell width, window dimension, half-width, sector dimension, parity period
$T_M(c) = (c_{\lvert r-r'\rvert})_{r,r'=0}^{M-1}$	symmetric Toeplitz matrix of the lag sequence c
$\Pi \in \mathbb{R}^{M\times h}$, $\Pi e_r = \frac{1}{\sqrt{2}}(e_r - e_{M-1-r})$	reflection-odd isometry, $\Pi^\mathsf{T}\Pi = I_h$
$A = \Pi^\mathsf{T} T_M(c)\Pi$, $A_{rr'} = c_{\lvert r-r'\rvert} - c_{M-1-r-r'}$	the odd window form
$a(\cdot)$; $S_D(s,u)$; ϱ_j, u_j	archimedean kernel; symmetrised hat; comb weights and node positions
L_P	odd compression of the lag sequence $(2,-1,0,\dots)$: tridiagonal $2I - E - E^\mathsf{T}$ with corner entry 3
$\mu_k = 4\sin^2(k\pi/N)$, $(t_k)_r = \frac{2}{\sqrt{N}}\sin\frac{2\pi k(r+1)}{N}$	parity eigenvalues and orthonormal parity modes, $k = 1,\dots,h$
$\widehat{a}_{ij} = t_i^\mathsf{T} A t_j$, $\widehat{A}_K = (\widehat{a}_{ij})_{i,j\le K}$	the raw Gram block
$\mathcal{M} = \operatorname{diag}(\mu_1,\dots,\mu_K)$, $G = \mathcal{M}^{-1/2}\widehat{A}_K\mathcal{M}^{-1/2}$	the normalised K-block
G_k, $g_k = e_1^\mathsf{T} G_k^{-1} e_1$, $s = \mu_1 t_1^\mathsf{T} A^{-1} t_1$	leading blocks, the ladder, the entry functional
x admissible: $x \in \mathbb{R}^K$, $x_1 = 1$	trial vector
$a_k = x_k/\sqrt{\mu_k}$, $v = \sum_{k\le K} a_k t_k$	coefficient vector and window vector
$Q(x) = x^\mathsf{T} G x = v^\mathsf{T} A v$	the value
$w_0 = \mathrm{Ac}_0(v)$, $w_d = 2\,\mathrm{Ac}_d(v) - \mathrm{Cv}_{M-1-d}(v)$ $(d\ge 1)$, $w_M := 0$	lag weights; Ac autocorrelation, Cv self-convolution
$(\Delta w)_d = w_d - w_{d+1}$, $\mathrm{TV}(x) = \lVert\Delta w\rVert_1$	lag increments and total variation
$C_d = \sum_{e\le d} c_e$, $C_{-1} := 0$	partial sums of the lag sequence
B, S, $r_{12}^2 = \widehat{a}_{12}^2/(\widehat{a}_{11}\widehat{a}_{22})$, $\nu_1 \ge \nu_2$	archimedean block, comb block, squared correlation, eigenvalues of $\widehat{A}_2$
$D(P,Q) = P_{11}Q_{22} + P_{22}Q_{11} - 2P_{12}Q_{12}$	polarisation of det on symmetric 2×2 matrices

Table 2: Notation, fixed for the whole paper.

1.4 Notation

Notation is fixed once, in Table 2, and not varied. Three choices need a word. The symbol Π denotes the reflection-odd isometry, so that B can be reserved for the archimedean Gram block of Section 6; $\mathcal{M}$ (calligraphic) denotes $\operatorname{diag}(\mu_1,\dots,\mu_K)$, so that M (upright) keeps its meaning as the window dimension; and the comb weights, written μ_j in [11], are written ϱ_j here, because μ_k is needed for the parity eigenvalues. The archimedean kernel is written $a(\cdot)$ and always carries its argument; the coefficient vector of a trial vector is written a and never does.

$X \succeq 0$ and $X \succ 0$ denote positive semidefiniteness and definiteness, $\preceq$ the Löwner order, $\lVert\cdot\rVert$ the Euclidean norm, e_k the k-th coordinate vector, $\lvert X\rvert$ the entrywise absolute value of a matrix, and sgn the entrywise sign. For $v \in \mathbb{R}^h$ the autocorrelation and self-convolution are

$$\mathrm{Ac}_d(v) = \sum_r v_r v_{r+d} \quad (d \ge 0), \qquad \mathrm{Cv}_j(v) = \sum_{r+r'=j} v_r v_{r'}, \tag{1}$$

both sums over indices in $\{0,\dots,h-1\}$ and empty sums being zero. *Admissible* always means $x_1 = 1$.

2 Setting

2.1 The odd window form and the comb class

We use the window form and the comb class of [11] verbatim and recall only what is needed. Fix a cell width $D > 0$ and an even M, put $\alpha = MD/2$, $h = M/2$, and let Π be the reflection-odd isometry of Table 2. Given a real lag sequence $c = (c_m)_{m=0}^{M-1}$ the odd window form is

$$A = \Pi^{\mathsf{T}} T_M(c)\, \Pi \in \mathbb{R}^{h\times h}, \qquad A_{rr'} = c_{|r-r'|} - c_{M-1-r-r'}. \tag{2}$$

The class we verify on is the comb class of [11, Def. 2.6]: with $a(\cdot)$ the archimedean kernel of that definition, a finite node set $\{(u_j, \varrho_j)\}_j$ with $u_j > 0$, $\varrho_j > 0$, and $S_D(s,u) = \frac{1}{2}[\Delta_D(s-u) + \Delta_D(s+u)]$ the symmetrised hat built from the unit-height triangle $\Delta_D(t) = \max\{0, 1 - |t|/D\}$ of half-width D,

$$c_m = a(mD) - \sum_{j:\, u_j \le 2\alpha} \varrho_j\, S_D(mD, u_j), \qquad m = 0, \ldots, M-1. \tag{3}$$

The *prime-power instance* is $u_j = \log n_j$ over the prime powers $n_j = p^\nu$ in increasing order with $\varrho_j = 2\Lambda(n_j)/\sqrt{n_j}$, Λ the von Mangoldt weight. By linearity of $c \mapsto A$ the split of (3) induces a split

$$A = A_{\text{arch}} - A_{\text{comb}}, \qquad A_{\text{comb}} = \sum_j \varrho_j\, A^{(j)}, \quad A^{(j)} = \Pi^{\mathsf{T}} T_M(S_D(\cdot D, u_j))\Pi. \tag{4}$$

Throughout, a *resolution* D is called *admissible* if $D < \min_j u_j$; for the prime-power instance this is implied by the admissibility convention of [11, Def. 5.1], which fixes D as a fraction of the smallest log-gap in the run.

Remark 2.1 (the arithmetic is not used)**.** Theorems 3.1, 4.1 and 6.3, and every lemma, proposition and corollary of Sections 2–6, hold for an *arbitrary* finite comb: the node positions u_j and the weights ϱ_j enter only as the data of (3), and no property of the sequence (u_j) or (ϱ_j) is used. Where the prime-power instance is mentioned it is as the instance the measurements of Sections 7–8 were taken on, and the only arithmetic facts consumed anywhere are the two classical gap inequalities used in [11] to check that a resolution is admissible [4, 20]. No arithmetic statement is claimed.

2.2 The parity spectrum

Let $c^L = (2, -1, 0, \ldots, 0)$ and let L_P be the odd window form (2) of c^L.

Lemma 2.2 (the parity compression and its spectrum)**.** *L_P is the tridiagonal matrix $2I - E - E^{\mathsf{T}}$ on $\mathbb{R}^h$ with corner entry $(L_P)_{h-1,h-1} = 3$; equivalently, L_P is the second-difference form with a Dirichlet condition at the outer end and the odd reflection $u_h = -u_{h-1}$ at the inner end. Its eigenpairs are*

$$\mu_k = 4\sin^2\!\Big(\frac{k\pi}{N}\Big), \qquad (t_k)_r = \frac{2}{\sqrt{N}} \sin\frac{2\pi k(r+1)}{N}, \qquad k = 1, \ldots, h,\ r = 0, \ldots, h-1, \tag{5}$$

with $N = 2h+1$, and $\{t_k\}_{k=1}^h$ is an orthonormal basis of $\mathbb{R}^h$. In particular $\mu_1 = 4\sin^2(\pi/N)$ is the smallest eigenvalue and

$$\frac{1}{\mu_1} \;\ge\; \frac{N^2}{4\pi^2} \;>\; \frac{h^2}{\pi^2}. \tag{6}$$

Proof. For c^L the Hankel part of (2) is $-c^L_{M-1-r-r'}$, which is nonzero only if $M-1-r-r' \in \{0,1\}$, i.e. only if $r + r' \in \{2h-2, 2h-1\}$; since $r, r' \le h-1$ the only possibility is $r = r' = h-1$, contributing $-c^L_1 = +1$. Hence L_P is tridiagonal with diagonal 2 except for the corner entry $2+1=3$, which is the first assertion. Reading the rows as a recurrence, row 0 is $2u_0 - u_1 = \mu u_0$, i.e. the convention $u_{-1}=0$, and row $h-1$ is $-u_{h-2}+3u_{h-1} = \mu u_{h-1}$, i.e. $-u_{h-2}+2u_{h-1}-u_h = \mu u_{h-1}$ with $u_h = -u_{h-1}$.

For the eigenpairs, put $u_r = \sin(\omega(r+1))$. Then $u_{-1}=0$ and the interior recurrence holds with $\mu = 2-2\cos\omega = 4\sin^2(\omega/2)$. The inner condition $u_h = -u_{h-1}$ reads $\sin(\omega(h+1)) + \sin(\omega h) = 2\sin(\frac{\omega N}{2})\cos(\frac{\omega}{2}) = 0$, and since $0<\omega<\pi$ the second factor is positive, so $\omega N/2 = k\pi$, i.e. $\omega = 2k\pi/N$ and $\mu = 4\sin^2(k\pi/N)$. The values $k=1,\dots,h$ give h distinct eigenvalues, hence all of them. For the normalisation, $\sum_{r=0}^{N-1}\sin^2(2\pi kr/N) = N/2$, the term $r=0$ vanishes, and the remaining $2h$ terms pair up as $r \leftrightarrow N-r$ with equal values, so $\sum_{r=1}^{h}\sin^2(2\pi kr/N) = N/4$ and the factor $2/\sqrt{N}$ normalises. Orthogonality is symmetry of L_P together with simplicity of the spectrum. Finally $\sin t \le t$ on $[0,\pi/2]$ gives $\mu_1 \le 4\pi^2/N^2$, which is (6); the last inequality is $N = 2h+1 > 2h$. □

The spectrum (5) is the classical Kac–Murdock–Szegő spectrum of the second difference in the reflection-odd sector [15], and μ_1 is its spectral gap. Everything in Sections 3 and 4 is driven by (6) and by nothing else.

2.3 The raw Gram block, the normalised block, and the ladder

Fix $K \le h$ (the verified surfaces use $K=16$; nothing below depends on the value). With $\{t_k\}$ of Lemma 2.2 put

$$\widehat{a}_{ij} = t_i^{\mathsf{T}} A t_j, \qquad \widehat{A}_K = (\widehat{a}_{ij})_{i,j\le K}, \qquad G = \mathcal{M}^{-1/2}\widehat{A}_K\mathcal{M}^{-1/2}, \qquad \mathcal{M} = \operatorname{diag}(\mu_1,\dots,\mu_K), \tag{7}$$

write G_k for the leading $k\times k$ block of G, and

$$g_k = e_1^{\mathsf{T}} G_k^{-1} e_1 \quad (k=1,\dots,K), \qquad s = \mu_1\, t_1^{\mathsf{T}} A^{-1} t_1. \tag{8}$$

The following three facts are classical; we record them in the form used later.

Lemma 2.3 (interlacing)**.** *Let $X \in \mathbb{R}^{n\times n}$ be symmetric and $P \in \mathbb{R}^{n\times m}$ an isometry. Then the eigenvalues of $P^{\mathsf{T}}XP$ interlace those of X; in particular $\lambda_{\min}(X) \le \lambda_{\min}(P^{\mathsf{T}}XP)$. Consequently a K-truncated Rayleigh minimum is an upper bound for $\lambda_{\min}$ and never a lower bound.*

Proof. Cauchy's interlacing theorem [3] in the Courant–Fischer form [5]. □

Lemma 2.4 (Schur nesting)**.** *Let $X \succ 0$ be partitioned as $\left[\begin{smallmatrix}\xi & b^{\mathsf{T}}\\ b & Y\end{smallmatrix}\right]$ with $\xi\in\mathbb{R}$. Then $Y\succ 0$ and*

$$\frac{1}{(X^{-1})_{11}} = \xi - b^{\mathsf{T}}Y^{-1}b, \qquad (X^{-1})_{11} = \frac{\det Y}{\det X}. \tag{9}$$

Proof. The first identity is the Schur complement of the $(1,1)$ pivot [21, 12]; the second is Jacobi's formula for a cofactor of the inverse, i.e. Cramer's rule. □

Lemma 2.5 (constrained minimum)**.** *Let $X\succ 0$ be $k\times k$. Then*

$$\min\{u^{\mathsf{T}}Xu:\ u\in\mathbb{R}^k,\ u_1=1\} = \frac{1}{e_1^{\mathsf{T}}X^{-1}e_1}, \tag{10}$$

attained exactly at $u^\star = X^{-1}e_1/(e_1^{\mathsf{T}}X^{-1}e_1)$. Equivalently, every *admissible u gives $u^{\mathsf{T}}Xu \ge 1/(e_1^{\mathsf{T}}X^{-1}e_1)$: an explicit trial vector is a certified lower bound for the reciprocal, never an upper one.*

Proof. Write $u = u^\star + \delta$ with $\delta_1 = 0$. Since $Xu^\star = e_1/(e_1^{\mathsf{T}}X^{-1}e_1)$, the cross term is $2\delta^{\mathsf{T}}Xu^\star = 2\delta_1/(e_1^{\mathsf{T}}X^{-1}e_1) = 0$, so $u^{\mathsf{T}}Xu = (u^\star)^{\mathsf{T}}Xu^\star + \delta^{\mathsf{T}}X\delta$ and $(u^\star)^{\mathsf{T}}Xu^\star = (u^\star)^{\mathsf{T}}e_1/(e_1^{\mathsf{T}}X^{-1}e_1) = 1/(e_1^{\mathsf{T}}X^{-1}e_1)$ because $u_1^\star = 1$. As $X\succ 0$ the remainder is positive unless $\delta = 0$. □

Proposition 2.6 (three forms of the ladder)**.** *Assume $G \succ 0$ and let $G = LL^\mathsf{T}$ be its Cholesky factorisation, $y = L^{-1}e_1$. Then for every $k \le K$:*

$$\textit{(prefix)} \qquad g_k = \sum_{j\le k} y_j^2; \tag{11}$$

$$\textit{(minor)} \qquad \frac{1}{g_k} = \frac{\det \widehat{A}_{[1..k]}}{\mu_1 \det \widehat{A}_{[2..k]}} \qquad (\det \widehat{A}_{[2..1]} := 1); \tag{12}$$

$$\textit{(regression)} \qquad \frac{g_k}{g_1} = \frac{1}{1-R_k^2}, \qquad R_k^2 = \frac{b^\mathsf{T} G_{HH}^{-1} b}{G_{11}}, \tag{13}$$

where in (13) the block G_k is partitioned as $\left[\begin{smallmatrix} G_{11} & b^\mathsf{T} \\ b & G_{HH}\end{smallmatrix}\right]$ with $H = \{2,\dots,k\}$. In particular $k \mapsto g_k$ is nondecreasing, strictly increasing whenever $y_k \ne 0$, and $R_k^2 \in [0,1)$ is the squared multiple correlation of the first coordinate on the remaining $k-1$.

Proof. Prefix. For a Cholesky factorisation the leading $k\times k$ block of G is $L_kL_k^\mathsf{T}$ with L_k the leading block of L, so $g_k = e_1^\mathsf{T} L_k^{-\mathsf{T}} L_k^{-1} e_1 = \|L_k^{-1}e_1\|^2$; and since L is lower triangular, forward substitution gives $L_k^{-1}e_1 = (L^{-1}e_1)_{1:k} = y_{1:k}$. Monotonicity is immediate.

Minor. By (9), $g_k = \det G_{[2..k]}/\det G_{[1..k]}$. Substituting $G = \mathcal{M}^{-1/2}\widehat{A}_K\mathcal{M}^{-1/2}$ gives $\det G_{[1..k]} = \det \widehat{A}_{[1..k]}/\prod_{j\le k}\mu_j$ and $\det G_{[2..k]} = \det \widehat{A}_{[2..k]}/\prod_{2\le j\le k}\mu_j$, and all factors cancel except μ_1, which yields (12).

Regression. By (9), $1/g_k = G_{11} - b^\mathsf{T} G_{HH}^{-1} b = G_{11}(1-R_k^2)$, while $g_1 = 1/G_{11}$; divide. ☐

Lemma 2.7 (diagonal invariance of the ladder gain)**.** *For every positive diagonal $\Lambda = \operatorname{diag}(d_1,\dots,d_K)$ the gain $g_k/g_1 = G_{11}\,(G_k^{-1})_{11}$ is unchanged under $G \mapsto \Lambda G\Lambda$.*

Proof. $(\Lambda G\Lambda)_{11} = d_1^2 G_{11}$ and $\big((\Lambda G\Lambda)_k^{-1}\big)_{11} = d_1^{-2}(G_k^{-1})_{11}$; the product is invariant. ☐

Lemma 2.7 localises the phenomenon: the gain does not see the $\mu^{-1/2}$ normalisation of (7) at all, so whatever makes g_K/g_1 large is a property of the raw Gram block $\widehat{A}_K$ alone, and the parity spectrum enters it only through the choice of the modes.

Proposition 2.8 (direction of the ladder)**.** *Assume $\widehat{A}_h \succ 0$, i.e. $A \succ 0$ on the parity sector. Then*

$$s \;\ge\; g_K \;\ge\; g_1 = \frac{1}{G_{11}}, \qquad \textit{equivalently} \qquad \frac{1}{s} \;\le\; \frac{1}{g_K} \;\le\; \frac{1}{g_1}, \tag{14}$$

with every ladder increment $g_{k+1} - g_k \ge 0$. The hypothesis is used only for the Cholesky factorisation of Proposition 2.6 and is, on the surfaces of Section 8, certified per window.

Proof. Let $T = [t_1,\dots,t_h]$, so T is orthogonal and $\widehat{A}_h = T^\mathsf{T} AT$. Then $A^{-1} = T\widehat{A}_h^{-1}T^\mathsf{T}$, hence $t_1^\mathsf{T} A^{-1} t_1 = (\widehat{A}_h^{-1})_{11}$; and with $\mathcal{M}_h = \operatorname{diag}(\mu_1,\dots,\mu_h)$ and $G^{(h)} = \mathcal{M}_h^{-1/2}\widehat{A}_h\mathcal{M}_h^{-1/2}$ one has $\big((G^{(h)})^{-1}\big)_{11} = \mu_1(\widehat{A}_h^{-1})_{11}$, i.e. $s = g_h$. Now apply (11): g_k is a partial sum of squares, so $g_1 \le g_K \le g_h = s$. ☐

2.4 Lag weights, the gauge law, and the boundary-free swap

Let $v \in \mathbb{R}^h$ and define the *lag weights* of v by

$$w_0 = \mathrm{Ac}_0(v), \qquad w_d = 2\,\mathrm{Ac}_d(v) - \mathrm{Cv}_{M-1-d}(v) \quad (1 \le d \le M-1), \qquad w_M := 0, \tag{15}$$

with Ac, Cv as in (1), and put $(\Delta w)_d = w_d - w_{d+1}$ and $\mathrm{TV} = \|\Delta w\|_1 = \sum_{d=0}^{M-1} |(\Delta w)_d|$. For an admissible x we write $w(x)$, $\mathrm{TV}(x)$ for the weights and total variation of the window vector $v = \sum_{k\le K} a_k t_k$, $a_k = x_k/\sqrt{\mu_k}$.

Lemma 2.9 (the weights represent the form; the gauge law). *For every $v \in \mathbb{R}^h$ and every lag sequence c,*

$$v^{\mathsf{T}} A v = \sum_{d=0}^{M-1} c_d\, w_d, \qquad w_0 = \|v\|^2, \qquad \sum_{d=0}^{M-1} w_d = 0. \tag{16}$$

In particular, for an admissible x, $Q(x) = x^{\mathsf{T}} G x = v^{\mathsf{T}} A v = \sum_d c_d w_d(x)$.

Proof. By (2), $v^{\mathsf{T}} A v = \sum_{r,r'} v_r v_{r'} c_{|r-r'|} - \sum_{r,r'} v_r v_{r'} c_{M-1-r-r'}$. The first sum collects the lag $d = |r-r'|$ with multiplicity 1 for $d = 0$ and 2 for $d \geq 1$, i.e. it equals $c_0 \operatorname{Ac}_0(v) + 2\sum_{d\geq 1} c_d \operatorname{Ac}_d(v)$. The second sum collects the index $d = M-1-r-r'$, and since $r, r' \leq h-1$ we have $r+r' \leq M-2$, so $d \geq 1$ throughout and the coefficient of c_d is $\operatorname{Cv}_{M-1-d}(v)$. This is the first identity of (16) with the weights (15), and it also shows $w_0 = \operatorname{Ac}_0(v) = \|v\|^2$: the reflection term cannot reach $d = 0$.

For the gauge law, sum (15). Over all $d \geq 0$, $\operatorname{Ac}_0(v) + 2\sum_{d\geq 1} \operatorname{Ac}_d(v) = \left(\sum_r v_r\right)^2$, and as d runs through $1, \ldots, M-1$ the index $j = M-1-d$ runs through $0, \ldots, M-2 = 2h-2$, so $\sum_{d\geq 1} \operatorname{Cv}_{M-1-d}(v) = \sum_{j=0}^{2h-2} \operatorname{Cv}_j(v) = \left(\sum_r v_r\right)^2$. The two contributions cancel. ☐

The gauge law has a consequence we use once: since $\sum_d w_d = 0$, the value $\sum_d c_d w_d$ is unchanged if a constant is added to every c_d. So the representation (16) determines the form only up to an additive normalisation of c, and any bound obtained from it may be optimised over that normalisation.

Lemma 2.10 (boundary-free summation by parts). *With $C_d = \sum_{e\leq d} c_e$, $C_{-1} = 0$, and $w_M = 0$,*

$$\sum_{d=0}^{M-1} c_d\, w_d \;=\; \sum_{d=0}^{M-1} C_d\, (\Delta w)_d \tag{17}$$

with no boundary term at either end.

Proof. Substituting $c_d = C_d - C_{d-1}$,

$$\sum_{d=0}^{M-1} c_d w_d = \sum_{d=0}^{M-1} C_d w_d - \sum_{d=0}^{M-1} C_{d-1} w_d = \sum_{d=0}^{M-1} C_d w_d - \sum_{d=0}^{M-2} C_d w_{d+1} = \sum_{d=0}^{M-1} C_d (w_d - w_{d+1}),$$

where the second equality shifted the index and used $C_{-1} = 0$, and the third used $w_M = 0$ to absorb the last term. ☐

This is Abel's summation by parts [1]. Both boundary terms vanish for structural reasons and not by a choice: $C_{-1} = 0$ by definition of a partial sum, and $w_M = 0$ because both terms of (15) vanish at $d = M$.

3 The total-variation floor

Theorem 3.1 (the total-variation floor). *Let $x \in \mathbb{R}^K$ be admissible, i.e. $x_1 = 1$, and let $w = w(x)$ be the lag weights of $v = \sum_{k\leq K} a_k t_k$, $a_k = x_k/\sqrt{\mu_k}$. Then*

$$\operatorname{TV}(x) \;\geq\; |w_0| \;=\; \|v\|^2 \;=\; \|a\|^2 \;\geq\; \frac{1}{\mu_1} \;=\; \frac{1}{4\sin^2(\pi/N)} \;\geq\; \frac{N^2}{4\pi^2}. \tag{18}$$

The bound involves no comb node, no taper and no truncation, and it holds for every lag sequence c — indeed the left- and right-hand sides do not depend on c at all.

Proof. Four steps.

(i) $w_0 = \|a\|^2$. By Lemma 2.9, $w_0 = \|v\|^2$; and $\|v\|^2 = \|a\|^2$ because $\{t_k\}$ is orthonormal (Lemma 2.2).

(ii) $\mathrm{TV} \geq |w_0|$. With $w_M = 0$ the increments telescope: $\sum_{d=0}^{M-1} (\Delta w)_d = w_0 - w_M = w_0$. Hence $|w_0| \leq \sum_{d=0}^{M-1} |(\Delta w)_d| = \mathrm{TV}$.

(iii) $\|a\|^2 \geq 1/\mu_1$. Admissibility gives $a_1 = x_1/\sqrt{\mu_1} = 1/\sqrt{\mu_1}$, so $\|a\|^2 \geq a_1^2 = 1/\mu_1$.

(iv) $1/\mu_1 \geq N^2/(4\pi^2)$. This is (6), i.e. $\sin t \leq t$. □

Three features of (18) are worth separating. It is a statement about the *normalisation*, not about the form: step (iii) is the only place where admissibility is used, and it is where the whole h^2 comes from. It is a statement about the *sector*: step (i) uses orthonormality of the parity modes and step (iv) uses their spectrum, both of which are the classical Kac–Murdock–Szegő data [15]. And it is *closed*: no constant is fitted, no window is excluded, and the right-hand side is an elementary function of h.

Corollary 3.2 (the price)**.** *For every admissible x,*

$$\frac{|Q(x)|}{\mathrm{TV}(x)} \;\leq\; \mu_1\, |Q(x)|. \tag{19}$$

Consequently, if a family of admissible trial vectors has values bounded by $|Q(x)| \leq U$ with U independent of h, the ratio it certifies is at most $U\mu_1 = O(h^{-2})$.

Proof. By Theorem 3.1, $\mathrm{TV}(x) \geq 1/\mu_1 > 0$, so $1/\mathrm{TV}(x) \leq \mu_1$; multiply by $|Q(x)| \geq 0$. The consequence is $\mu_1 = 4\sin^2(\pi/N) \leq 4\pi^2/N^2$. □

Corollary 3.2 is the load-bearing statement of the section. It says that the pair (value, total variation) cannot be improved by choosing a better trial vector: the ratio is pinned by the value alone, up to the spectral gap. Any route whose output is the ratio therefore pays a factor h^{-2} that is a property of the parity spectrum meeting the normalisation, and not a property of the comb.

Corollary 3.3 (what summation by parts can certify)**.** *For every admissible x,*

$$|Q(x)| \;\leq\; \Big(\max_{0\leq d\leq M-1} |C_d|\Big)\ \mathrm{TV}(x), \qquad \textit{hence} \qquad \max_d |C_d| \;\geq\; \frac{|Q(x)|}{\mathrm{TV}(x)}. \tag{20}$$

Combined with (19)*, a family of bounded value certifies* $\max_d |C_d| \geq$ *a quantity that is $O(h^{-2})$, and nothing stronger. By the gauge law of Lemma 2.9 the estimate may be applied to $c + \gamma$ for any $\gamma \in \mathbb{R}$, so* $\max_d |C_d|$ *may be replaced by* $\inf_\gamma \max_d |C_d + \gamma(d+1)|$.

Proof. Lemma 2.10 and Hölder: $|\sum_d C_d (\Delta w)_d| \leq \max_d |C_d| \cdot \|\Delta w\|_1$. The final sentence is the remark after Lemma 2.9, since replacing c_d by $c_d + \gamma$ leaves Q unchanged and replaces C_d by $C_d + \gamma(d+1)$. □

Corollary 3.4 (the floor at one coordinate)**.** *For every $v \in \mathbb{R}^h \setminus \{0\}$ with $t_1^\mathsf{T} v \neq 0$,*

$$\frac{\mathrm{TV}(v)}{(t_1^\mathsf{T} v)^2} \;\geq\; 1, \qquad \textit{because} \qquad \mathrm{TV}(v) \;\geq\; |w_0| = \|v\|^2 \;\geq\; (t_1^\mathsf{T} v)^2. \tag{21}$$

No normalisation is required: (21) *holds for every window vector, and for admissible x it reduces to* (18) *because* $(t_1^\mathsf{T} v)^2 = a_1^2 = 1/\mu_1$.

Proof. The first two relations are steps (i)–(ii) of Theorem 3.1, which used no admissibility; the third is Cauchy–Schwarz with $\|t_1\| = 1$. □

Remark 3.5 (negative control: the floor is a property of the normalisation)**.** Step (iii) is the only step that uses $x_1 = 1$, and it is indispensable: for $\lambda \neq 0$ the vector λx has $\mathrm{TV}(\lambda x) = \lambda^2\,\mathrm{TV}(x)$ (both Ac and Cv are quadratic in v, and v is linear in x), so $\mu_1\,\mathrm{TV}(\lambda x) = \lambda^2 \mu_1\,\mathrm{TV}(x)$ falls below 1 as soon as $|\lambda|$ is small enough. An inadmissible vector therefore undercuts the floor, and recognising it as inadmissible is the whole content of the observation. This is also the reason Theorem 4.1 below is a statement about rays.

Measurement 3.6 (sharpness of the floor)**.** On Surface II of Section 8 — 27 windows of the frame-A family, $h = 142$ to 1445, all with $\mathrm{cond}(G) \leq 10^{12}$ — the floor of Theorem 3.1 is tight to about one order of magnitude: the measured ratio is

$$\mu_1\ \mathrm{TV}(x) \in [5.00,\ 23.20] \tag{22}$$

at the optimiser of the declared trial family, while the floor itself takes the values $2.06 \cdot 10^3$ to $1.77 \cdot 10^5$, growing as $h^{+1.997}$ (a least-squares fit, reported for orientation and not used anywhere) against the closed h^2 of (6). The inadmissible control of Remark 3.5 returns $\mu_1\,\mathrm{TV} \approx 7 \cdot 10^{-4}$ at $\lambda = 10^{-2}$, i.e. it undercuts the floor by the predicted factor λ^2. The trial family is the declared tapered mode family also used in Measurement 4.5, its taper width running over a fixed geometric list; the caps of Surface II are written out in Section 8.2.

We close the section by saying plainly what the floor does not know about. It does not know where the comb nodes are, or how many there are, or what their weights are; it does not know whether the trial vector was tapered, and it does not know at which K the mode family was truncated. It is a statement about $\{t_k\}$, about μ_1, and about the normalisation $x_1 = 1$. The next section shows that the last of these three cannot be traded away.

4 Gauge rigidity of the normalisation

Theorem 3.1 raises an obvious question: the floor is $1/\mu_1$ because the normalisation is imposed on the mode with eigenvalue μ_1, so could a different *entry spectrum* — a reweighted, shifted or otherwise reparametrised family — produce a flatter floor and hence a better ratio? The answer is no, and it is no for a structural reason.

Theorem 4.1 (the normalisation is a gauge)**.** *For $a \in \mathbb{R}^K$ let $v(a) = \sum_{k \leq K} a_k t_k$ and*

$$\widetilde{Q}(a) = v(a)^{\mathsf{T}} A v(a), \qquad \widetilde{\mathrm{TV}}(a) = \|\Delta w(v(a))\|_1. \tag{23}$$

Then:

(i) *$\widetilde{Q}$ and $\widetilde{\mathrm{TV}}$ are homogeneous of degree 2: $\widetilde{Q}(\lambda a) = \lambda^2 \widetilde{Q}(a)$ and $\widetilde{\mathrm{TV}}(\lambda a) = \lambda^2 \widetilde{\mathrm{TV}}(a)$ for all $\lambda \in \mathbb{R}$.*

(ii) *Hence the ratio $\varrho(a) = \widetilde{Q}(a)/\widetilde{\mathrm{TV}}(a)$ is constant on rays: it is a function of the direction of a alone.*

(iii) *Let $\mu' = (\mu'_1, \ldots, \mu'_K)$ be* any *positive entry spectrum and let $x \mapsto a$, $a_k = x_k/\sqrt{\mu'_k}$, be the associated parametrisation. Then the constraint $x_1 = 1$ describes the affine slice $\{a : a_1 = 1/\sqrt{\mu'_1}\}$, every ray $\{\lambda a : \lambda > 0\}$ with $a_1 > 0$ meets that slice exactly once, and consequently*

$$\{\varrho(a) : a_k = x_k/\sqrt{\mu'_k},\ x_1 = 1\} \ = \ \{\varrho(a) : a \in \mathbb{R}^K,\ a_1 > 0\} \tag{24}$$

independently of μ'*. The entry normalisation is a gauge: the range of the ratio is the same in every sector.*

Proof. (i) $a \mapsto v(a)$ is linear, and by (1) both Ac_d and Cv_j are quadratic forms in v, so $w(v(\lambda a)) = \lambda^2 w(v(a))$ by (15), hence Δw and its ℓ^1 norm scale by λ^2; and $\widetilde{Q}$ is a quadratic form in a.

(ii) Immediate from (i), both numerator and denominator having the same degree.

(iii) The map $x \mapsto a$ is a linear bijection of $\mathbb{R}^K$ taking $\{x_1 = 1\}$ onto $\{a_1 = 1/\sqrt{\mu'_1}\}$. A ray $\{\lambda a : \lambda > 0\}$ with $a_1 > 0$ meets $\{a_1 = \theta\}$ exactly once for every $\theta > 0$, namely at $\lambda = \theta/a_1$. So the slice $\{a_1 = 1/\sqrt{\mu'_1}\}$ is a cross-section of the family of all rays with $a_1 > 0$, whatever the value of $\mu'_1 > 0$; by (ii) the ratio takes the same set of values on the slice as on the family of rays, which is (24). The right-hand side does not mention μ'. □

Theorem 4.1 is a fact about degree-two homogeneous functionals, and we state it modestly for that reason. Its force is entirely in the consequence: since the accessible range of the ratio is a sector-independent set, and since Theorem 3.1 bounds that range in the parity parametrisation, no change of sector can enlarge it.

Corollary 4.2 (no sector change removes the h^2)**.** *Let μ' be any positive entry spectrum and let x' be admissible for it. Then*

$$\frac{|\widetilde{Q}(a')|}{\widetilde{\mathrm{TV}}(a')} \;\le\; \mu_1 \,\sup\{|\widetilde{Q}(a)| : a \textit{ in the parity slice}\}, \tag{25}$$

with $\mu_1 = 4\sin^2(\pi/N)$ the parity gap — i.e. the bound of Corollary 3.2 transfers verbatim, with the parity *gap, to every sector. In particular a family of trial vectors with h-bounded value in some sector certifies a ratio $O(h^{-2})$ in that sector as well.*

Proof. By (24) the value $\varrho(a')$ is attained by some a in the parity slice ($\mu' = \mu$), i.e. by some admissible x; apply Corollary 3.2 to that x. □

The next proposition is the quantitative form of Corollary 4.2 for the simplest reparametrisation, a rigid shift of the entry spectrum. It is a two-line computation, and we state it as a proposition rather than dressing it up: the point is not that the computation is hard but that the product it computes is *exactly* constant.

Proposition 4.3 (floor × transfer is conserved)**.** *Let $\tau \ge 0$ and let $\mu'_k = \mu_k + \tau$ be the shifted entry spectrum, so that admissibility in the shifted sector reads $a'_1 = 1/\sqrt{\mu_1 + \tau}$. Then the shifted floor of Theorem 3.1 is $\widetilde{\mathrm{TV}}(a') \ge 1/(\mu_1 + \tau)$, the transfer factor relating the two normalisations is*

$$\theta = \frac{(a'_1)^2}{a_1^2} = \frac{\mu_1}{\mu_1 + \tau}, \qquad \frac{1}{\theta} = 1 + \frac{\tau}{\mu_1}, \tag{26}$$

and

$$\underbrace{\frac{1}{\mu_1 + \tau}}_{\textit{shifted floor}} \cdot \underbrace{\Big(1 + \frac{\tau}{\mu_1}\Big)}_{\textit{transfer}} = \frac{1}{\mu_1 + \tau} \cdot \frac{\mu_1 + \tau}{\mu_1} = \frac{1}{\mu_1} \qquad \textit{identically, at every shift } \tau. \tag{27}$$

A shift does flatten the floor; it does not remove the h^2, which lives in the ratio and not in the floor.

Proof. The floor is Theorem 3.1 applied in the shifted parametrisation, whose step (iii) reads $\|a'\|^2 \ge (a'_1)^2 = 1/(\mu_1 + \tau)$. For the transfer, the two admissibility constraints differ only in the value of the first coefficient, $a_1 = 1/\sqrt{\mu_1}$ against $a'_1 = 1/\sqrt{\mu_1 + \tau}$; by Theorem 4.1(i) a value computed at a' is converted to the corresponding value on the same ray in the unshifted slice by the factor $(a_1/a'_1)^2 = 1/\theta$. The displayed product is then (27). □

Proposition 4.4 (the null mode, and the other sector)**.** *Let* $c^L = (2, -1, 0, \dots)$ *be as in Lemma 2.2.*

(i) *The parity family* $\mu_k = 4\sin^2(k\pi/N)$, $k = 1, \dots, h$, *is the restriction to* $k \ge 1$ *of the sine family* $\{4\sin^2(k\pi/N)\}_{k\ge 0}$, *whose member at* $k = 0$ *is* $\mu_0 = 0$.

(ii) *The parametrisation* $a_k = x_k/\sqrt{\mu'_k}$ *of Theorem 4.1*(iii) *requires* $\mu'_k > 0$. *If a mode family containing a null mode is used and the normalisation is imposed on that mode, the constraint has* no *representative* a *of finite norm; if the normalisation is imposed elsewhere, step (iii) of Theorem 3.1 yields the floor of the normalised mode and the null mode contributes nothing. In neither case is a flatter admissible floor obtained.*

(iii) *Write* $\Pi_+ e_r = \frac{1}{\sqrt{2}}(e_r + e_{M-1-r})$ *for the reflection-even isometry. The complementary compression* $\Pi_+^\mathsf{T} T_M(c^L)\Pi_+$ *has spectrum* $\{4\sin^2\big((2k+1)\pi/(2N)\big)\}_{k=0}^{h-1}$, *whose minimum* $4\sin^2(\pi/(2N))$ *is* strictly smaller *than* μ_1. *Hence no reflection sector of the second-difference form has a lowest eigenvalue exceeding* μ_1.

Proof. (i) is the formula of Lemma 2.2 read at $k = 0$.

(ii) is the definition of the parametrisation together with step (iii) of Theorem 3.1, which bounds $\|a\|^2$ below by the square of the *normalised* coordinate only.

(iii) For c^L the even compression is $(\Pi_+^\mathsf{T} T_M(c^L)\Pi_+)_{rr'} = c^L_{|r-r'|} + c^L_{M-1-r-r'}$, which by the computation in the proof of Lemma 2.2 is tridiagonal $2I - E - E^\mathsf{T}$ with corner entry $2 - 1 = 1$; as a recurrence this is $u_{-1} = 0$ and $u_h = +u_{h-1}$. With $u_r = \sin(\omega(r+1))$ the inner condition reads $\sin(\omega(h+1)) - \sin(\omega h) = 2\cos(\frac{\omega N}{2})\sin(\frac{\omega}{2}) = 0$, hence $\omega N/2 = (k + \frac{1}{2})\pi$ and $\mu = 4\sin^2\big((2k+1)\pi/(2N)\big)$ for $k = 0, \dots, h-1$, which are h distinct values, hence all. The minimum is at $k = 0$ and $4\sin^2(\pi/(2N)) < 4\sin^2(\pi/N) = \mu_1$ because sin is increasing on $[0, \pi/2]$. $\square$

Part (iii) is the precise sense in which the parity compression is the best of the two reflection sectors for this purpose, and part (ii) is the sense in which enlarging the mode family removes the statement instead of improving it. Together with Theorem 4.1 and Proposition 4.3 this closes the sector question: reweighting, shifting, or enlarging the entry spectrum changes neither the accessible range of the ratio nor the conserved product.

Measurement 4.5 (the sector battery)**.** Theorem 4.1 and Proposition 4.3 are proved statements; the following is their machine verification on a declared battery, reported because it also checks the implementation of the transport map. Over 35 weight families $\times$ 6 taper widths $\times$ 27 windows of Surface II the ratio ϱ is reproduced across sectors with residual at most $8.8 \cdot 10^{-10}$ against a declared round-off horizon of $2.6 \cdot 10^{-8}$, and the product (27) is reproduced as an exact identity at every shift of the declared shift list. Transporting the x-coordinates instead of the ray — the wrong map — moves the ratio by a large factor on the spread-out weight families, which is the mutation control of the battery.

5 The two-dimensional reduction

Sections 3 and 4 priced the trial-vector route. This section identifies what is left: at $K = 2$ the optimal trial vector is available in closed form, so the entire question collapses onto a single scalar, the squared correlation of the two lowest parity modes in the metric of A.

Proposition 5.1 (the closed vector is exact at $K = 2$)**.** *Assume* $G_2 \succ 0$ *and put* $r_{12}^2 = G_{12}^2/(G_{11}G_{22}) = \widehat{a}_{12}^2/(\widehat{a}_{11}\widehat{a}_{22})$. *Then* $u = (1, -G_{21}/G_{22})$ *attains the constrained minimum of Lemma 2.5 at* $k = 2$*:*

$$u^\mathsf{T} G_2 u = \frac{1}{g_2} = G_{11}(1 - r_{12}^2), \qquad G_{11}\, g_2 = \frac{1}{1 - r_{12}^2} \quad \textit{identically.} \tag{28}$$

Proof. Since $G_2 \succ 0$ we have $G_{11} > 0$, $G_{22} > 0$ and $\det G_2 > 0$.

The minimum. Lemma 2.4 applied to $X = G_2$ with $\xi = G_{11}$, $b = G_{12}$ and $Y = G_{22}$ gives

$$\frac{1}{g_2} = \frac{1}{(G_2^{-1})_{11}} = G_{11} - \frac{G_{12}^2}{G_{22}} = G_{11}\Big(1 - \frac{G_{12}^2}{G_{11}G_{22}}\Big) = G_{11}(1 - r_{12}^2),$$

and by Lemma 2.5 this number is the constrained minimum.

The minimiser. By Cramer's rule $G_2^{-1}e_1 = (\det G_2)^{-1}(G_{22}, -G_{12})^\mathsf{T}$, so $G_2^{-1}e_1/(e_1^\mathsf{T}G_2^{-1}e_1) = (1, -G_{12}/G_{22})^\mathsf{T} = u$ because $G_{21} = G_{12}$; Lemma 2.5 identifies this vector as the unique minimiser. It is worth verifying the value on u by substitution rather than through the inverse:

$$u^\mathsf{T}G_2u = G_{11} - 2\frac{G_{12}^2}{G_{22}} + \frac{G_{12}^2}{G_{22}^2}\,G_{22} = G_{11} - \frac{G_{12}^2}{G_{22}},$$

the same number, and this second computation uses only $G_{22} \neq 0$.

The gain. $g_1 = 1/G_{11}$, so dividing the first display by g_1 gives $G_{11}g_2 = 1/(1 - r_{12}^2)$, which is the case $k = 2$ of (13) with $R_2^2 = r_{12}^2$.

The two readings of r_{12}^2 agree. By (7), $G_{ij} = \widehat{a}_{ij}/\sqrt{\mu_i\mu_j}$, hence

$$\frac{G_{12}^2}{G_{11}G_{22}} = \frac{\widehat{a}_{12}^2/(\mu_1\mu_2)}{(\widehat{a}_{11}/\mu_1)\,(\widehat{a}_{22}/\mu_2)} = \frac{\widehat{a}_{12}^2}{\widehat{a}_{11}\widehat{a}_{22}}\ :$$

the diagonal congruence cancels, as it must by Lemma 2.7. □

Proposition 5.2 (pivot identity)**.** *Assume $G_k \succ 0$ and let $x^\star$ be the minimiser of Lemma 2.5. Then for every δ with $\delta_1 = 0$,*

$$(x^\star + \delta)^\mathsf{T}G_k(x^\star + \delta) = \frac{1}{g_k} + \delta^\mathsf{T}G_k\delta \qquad \textit{exactly,} \tag{29}$$

so the relative excess $\rho = g_k\,\delta^\mathsf{T}G_k\delta$ of any explicit candidate is an exact number and never an estimate.

Proof. Write $x^\star = G_k^{-1}e_1/g_k$, so that $G_kx^\star = e_1/g_k$ and $x_1^\star = e_1^\mathsf{T}G_k^{-1}e_1/g_k = 1$. Expanding the quadratic form,

$$(x^\star + \delta)^\mathsf{T}G_k(x^\star + \delta) = (x^\star)^\mathsf{T}G_kx^\star + 2\,\delta^\mathsf{T}G_kx^\star + \delta^\mathsf{T}G_k\delta.$$

The first term is $(x^\star)^\mathsf{T}e_1/g_k = x_1^\star/g_k = 1/g_k$ and the cross term is $2\delta^\mathsf{T}e_1/g_k = 2\delta_1/g_k = 0$. No inequality is used at any point, which is the content of the word "exactly" in (29): $\delta^\mathsf{T}G_k\delta$ is not an estimate for the gap between an explicit candidate and the optimum, it *is* the gap. □

Remark 5.3 (the sign of the closed vector is not cosmetic)**.** At $k = 2$ the candidate $(1, +G_{12}/G_{22})$ differs from the minimiser of Proposition 5.1 by $\delta = (0, 2G_{12}/G_{22})$, and (29) prices it exactly: $\delta^\mathsf{T}G_2\delta = 4G_{12}^2/G_{22}$, so the relative excess is

$$\rho = g_2\,\delta^\mathsf{T}G_2\delta = \frac{4G_{12}^2/G_{22}}{G_{11}(1 - r_{12}^2)} = \frac{4r_{12}^2}{1 - r_{12}^2} \qquad \text{identically,} \tag{30}$$

which diverges precisely where the cancellation is deep. The pivot identity therefore also quantifies how a wrong closed vector fails, and it does so without a single inequality.

Proposition 5.4 (the entrywise perturbation, and one cancellation ratio)**.** *Let $x^\star$ be as above, $\sigma = \operatorname{sgn}(x^\star)$, $\Theta_k = |x^\star|^\mathsf{T}|G_k|\,|x^\star|$, and for $\varepsilon > 0$ let E be the sign-aligned entrywise perturbation $E_{ij} = \varepsilon|G_{ij}|\sigma_i\sigma_j$. Then*

$$\big|(x^\star)^\mathsf{T}Ex^\star\big| = \varepsilon\,\Theta_k \qquad \textit{exactly,} \tag{31}$$

so the entry threshold at which the perturbation exhausts the value is $\varepsilon_{\mathrm{ent}}(k) = \rho\,(1/g_k)/\Theta_k$ *with* ρ *of* (29)*. At* $k = 2$ *the ratio is closed:*

$$\Theta_2 = G_{11}(1 + 3r_{12}^2), \qquad \frac{1/g_2}{\Theta_2} = \frac{1 - r_{12}^2}{1 + 3r_{12}^2}, \tag{32}$$

and $1 + 3r_{12}^2 \to 4$ *as* $r_{12}^2 \to 1$.

Proof. The union identity. Since $x_i^\star \sigma_i = |x_i^\star|$ for every i,

$$(x^\star)^\mathsf{T} E x^\star = \varepsilon \sum_{i,j} x_i^\star \sigma_i \, |G_{ij}| \, \sigma_j x_j^\star = \varepsilon \sum_{i,j} |x_i^\star| \, |G_{ij}| \, |x_j^\star| = \varepsilon \, \Theta_k,$$

and every term of the middle sum is nonnegative, so no cancellation occurs and the modulus may be taken termwise: the sign-aligned perturbation is the one that realises the ℓ^1 mass of G_k in the direction $x^\star$. For the threshold, (29) says the value at $x^\star$ is $1/g_k$ and that a relative excess ρ is an absolute excess ρ/g_k, so the perturbation exhausts the value exactly when $\varepsilon\Theta_k = \rho/g_k$.

The closed ratio at $k = 2$. By Proposition 5.1, $x^\star = (1, -G_{12}/G_{22})$, hence $|x^\star| = (1, |G_{12}|/G_{22})$ because $G_{22} > 0$; and since $G_{11}, G_{22} > 0$,

$$\Theta_2 = \underbrace{G_{11}}_{(1,1)} + \underbrace{2\,\frac{|G_{12}|}{G_{22}}\,|G_{12}|}_{(1,2)\text{ and }(2,1)} + \underbrace{\frac{G_{12}^2}{G_{22}^2}\,G_{22}}_{(2,2)} = G_{11} + 3\,\frac{G_{12}^2}{G_{22}} = G_{11}(1 + 3r_{12}^2).$$

The constant is 3 and not 2 because the off-diagonal is counted twice and the $(2,2)$ entry once. Dividing (28) by this gives (32), and $1 + 3r_{12}^2 \to 4$ as $r_{12}^2 \to 1$. ☐

Remark 5.5. Proposition 5.4 is the reason the three natural readings of the same obstruction — a determinant ratio, the scalar $1 - r_{12}^2$, and an entrywise perturbation threshold — are one object rather than three: at $k = 2$ they differ by the explicit factor $1 + 3r_{12}^2 \in [1, 4]$. We state the two formulae and leave the reading to the reader; "three dresses of one inequality" is an observation about identities, not a theorem.

Lemma 5.6 (Wronskian telescope)**.** *Let* $u = t_1$*,* $v = t_2$ *be the two lowest parity modes of Lemma 2.2, extended by* $u_{-1} = v_{-1} = 0$ *and* $u_h = -u_{h-1}$*,* $v_h = -v_{h-1}$*, and put* $W_r = u_{r+1}v_r - u_r v_{r+1}$*. Then*

$$(\mu_1 - \mu_2)\, u_r v_r = W_{r-1} - W_r \quad (0 \le r \le h-1), \qquad W_{-1} = W_{h-1} = 0, \tag{33}$$

and summing (33) *re-derives* $t_1^\mathsf{T} t_2 = 0$.

Proof. By the definition of W_r,

$$W_{r-1} - W_r = (u_r v_{r-1} - u_{r-1} v_r) - (u_{r+1} v_r - u_r v_{r+1}) = u_r(v_{r-1} + v_{r+1}) - v_r(u_{r-1} + u_{r+1}).$$

Rows $0, \ldots, h-1$ of $L_P t_1 = \mu_1 t_1$ read $-u_{r-1} + 2u_r - u_{r+1} = \mu_1 u_r$ under the two conventions $u_{-1} = 0$ and $u_h = -u_{h-1}$ — this is exactly the reading of the rows carried out in the proof of Lemma 2.2 — i.e. $u_{r-1} + u_{r+1} = (2 - \mu_1)u_r$, and likewise $v_{r-1} + v_{r+1} = (2 - \mu_2)v_r$. Substituting,

$$W_{r-1} - W_r = (2 - \mu_2)u_r v_r - (2 - \mu_1)u_r v_r = (\mu_1 - \mu_2)\, u_r v_r,$$

which is (33). The two ends are the two boundary conventions and nothing else: $W_{-1} = u_0 v_{-1} - u_{-1} v_0 = 0$ is the Dirichlet end, and

$$W_{h-1} = u_h v_{h-1} - u_{h-1} v_h = (-u_{h-1})v_{h-1} - u_{h-1}(-v_{h-1}) = 0$$

is the odd reflection, which by the computation in the proof of Lemma 2.2 is precisely the corner entry 3 of L_P. Summing (33) over $r = 0, \ldots, h-1$ telescopes the right-hand side to $W_{-1} - W_{h-1} = 0$, so $(\mu_1 - \mu_2)\, t_1^\mathsf{T} t_2 = 0$; as $\mu_1 \neq \mu_2$, orthogonality of the two lowest parity modes is a *consequence* of the telescope rather than an independent input. ☐

Lemma 5.7 (maximal minors carry no arithmetic). *With $u = t_1/\sqrt{\mu_1}$, $v = t_2/\sqrt{\mu_2}$ and $\mathcal{W}_{rr'} = u_r v_{r'} - u_{r'} v_r$,*

$$\tfrac{1}{2}\sum_{r,r'} \mathcal{W}_{rr'}^2 = \|u\|^2\|v\|^2 - \langle u, v\rangle^2 = \frac{1}{\mu_1\mu_2} \qquad \textit{exactly.} \tag{34}$$

Proof. Expanding the square and using the symmetry of the double sum,

$$\tfrac{1}{2}\sum_{r,r'} \big(u_r v_{r'} - u_{r'} v_r\big)^2 = \tfrac{1}{2}\Big[\sum_{r,r'} u_r^2 v_{r'}^2 - 2\sum_{r,r'} u_r v_r\, u_{r'} v_{r'} + \sum_{r,r'} u_{r'}^2 v_r^2\Big] = \|u\|^2\|v\|^2 - \langle u, v\rangle^2,$$

the two outer sums being equal; this is Lagrange's identity [17]. For the value, t_1 and t_2 are orthonormal by Lemma 2.2, so $\|u\|^2 = 1/\mu_1$, $\|v\|^2 = 1/\mu_2$ and $\langle u, v\rangle = 0$. □

Lemma 5.7 is the reason the near-degeneracy cannot be attributed to the mode pair: the full Gram determinant of the two rescaled modes is the closed number $1/(\mu_1\mu_2)$, with no comb in it. Whatever becomes small in Problem 7.1 becomes small in the metric of A and nowhere else.

Lemma 5.8 (the determinant reading, free of positivity). *With $\nu_1 \geq \nu_2$ the eigenvalues of $\widehat{A}_2$,*

$$1 - r_{12}^2 = \frac{\det \widehat{A}_2}{\widehat{a}_{11}\widehat{a}_{22}} = \frac{\nu_1\nu_2}{\widehat{a}_{11}\widehat{a}_{22}} = \frac{\det G_2}{G_{11}G_{22}}, \tag{35}$$

and no positivity of A, of $\widehat{A}_2$ or of G is used.

Proof. $\det \widehat{A}_2 = \widehat{a}_{11}\widehat{a}_{22} - \widehat{a}_{12}^2$, so dividing by $\widehat{a}_{11}\widehat{a}_{22}$ gives the first equality by the definition of r_{12}^2. The second is $\det \widehat{A}_2 = \nu_1\nu_2$, valid for every symmetric 2×2 matrix whatever the signs of its eigenvalues. The third repeats the first computation for $G_2 = \mathcal{M}_2^{-1/2}\widehat{A}_2\mathcal{M}_2^{-1/2}$, where the diagonal factors cancel exactly as in the last step of the proof of Proposition 5.1. The chain uses only $\widehat{a}_{11}\widehat{a}_{22} \neq 0$: no eigenvalue is assumed positive, no factorisation is performed, and ν_2 may be negative on both sides of every equality. □

Lemma 5.8 is load-bearing in a negative sense, and we say so plainly: it is what makes the whole reduction free of any positivity input about A. Whether A is positive definite on the parity sector is, on the surfaces of Section 8, a numerical fact about finitely many finite matrices; no statement of this paper is routed through a uniform version of it.

Lemma 5.9 (an unconditional bound for the large eigenvalue). $\nu_1 \leq \max(\widehat{a}_{11}, \widehat{a}_{22}) + |\widehat{a}_{12}|$, *closed in the three entries and uniform in the window.*

Proof. This is Gershgorin's theorem [9] for $\widehat{A}_2$; we give the two lines because the statement is used with no positivity hypothesis anywhere. The eigenvalues of a symmetric 2×2 matrix are

$$\nu_{1,2} = \tfrac{1}{2}\Big[\widehat{a}_{11} + \widehat{a}_{22} \pm \sqrt{(\widehat{a}_{11} - \widehat{a}_{22})^2 + 4\widehat{a}_{12}^2}\Big],$$

and $\sqrt{p^2 + q^2} \leq |p| + |q|$ gives $\sqrt{(\widehat{a}_{11} - \widehat{a}_{22})^2 + 4\widehat{a}_{12}^2} \leq |\widehat{a}_{11} - \widehat{a}_{22}| + 2|\widehat{a}_{12}|$, whence

$$\nu_1 \leq \tfrac{1}{2}\big[\widehat{a}_{11} + \widehat{a}_{22} + |\widehat{a}_{11} - \widehat{a}_{22}|\big] + |\widehat{a}_{12}| = \max(\widehat{a}_{11}, \widehat{a}_{22}) + |\widehat{a}_{12}|.$$

The closed form makes visible in what sense the bound is uniform in the window: it is a function of the three entries alone, with no constant fitted and no window excluded. □

Proposition 5.10 (the determinant returns: the $t^\star$ collapse)**.** *Assume $\widehat{a}_{11}, \widehat{a}_{22} > 0$ and consider the one-parameter family $u(t) = (t, -1)$. At $t^\star = \sqrt{\widehat{a}_{22}/\widehat{a}_{11}}$ the Rayleigh quotient of $\widehat{A}_2$ is*

$$R(t^\star) = \frac{u(t^\star)^\mathsf{T} \widehat{A}_2 u(t^\star)}{\|u(t^\star)\|^2} = \frac{2\sqrt{\widehat{a}_{11}\widehat{a}_{22}}\ \det \widehat{A}_2}{(\widehat{a}_{11} + \widehat{a}_{22})\big(\sqrt{\widehat{a}_{11}\widehat{a}_{22}} + \widehat{a}_{12}\big)} \qquad \textit{identically,} \tag{36}$$

so the only member of the family that meets the threshold does so by reintroducing $\det \widehat{A}_2$*. If in addition $\widehat{A}_2 \succ 0$, then*

$$\nu_2 \;\le\; \frac{2 \det \widehat{A}_2}{\widehat{a}_{11} + \widehat{a}_{22}} \;=\; \frac{2\nu_1\nu_2}{\nu_1 + \nu_2} \;\le\; 2\nu_2, \tag{37}$$

i.e. the harmonic mean of the two eigenvalues bounds ν_2 from above and is tight within a factor 2*.*

Proof. The closed form, unconditional. Put $\tau = t^\star = \sqrt{\widehat{a}_{22}/\widehat{a}_{11}}$, well defined because $\widehat{a}_{11}, \widehat{a}_{22} > 0$, and note the two identities $\tau^2\widehat{a}_{11} = \widehat{a}_{22}$ and $\tau\widehat{a}_{11} = \sqrt{\widehat{a}_{11}\widehat{a}_{22}}$, the second of which also gives $\widehat{a}_{22} = \tau\sqrt{\widehat{a}_{11}\widehat{a}_{22}}$. The numerator of the Rayleigh quotient is

$$u(\tau)^\mathsf{T} \widehat{A}_2 u(\tau) = \tau^2\widehat{a}_{11} - 2\tau\widehat{a}_{12} + \widehat{a}_{22} = 2\big(\widehat{a}_{22} - \tau\widehat{a}_{12}\big) = 2\tau\big(\sqrt{\widehat{a}_{11}\widehat{a}_{22}} - \widehat{a}_{12}\big),$$

and the denominator is $\|u(\tau)\|^2 = \tau^2 + 1 = (\widehat{a}_{11} + \widehat{a}_{22})/\widehat{a}_{11}$. Hence

$$R(t^\star) = \frac{2\tau\widehat{a}_{11}\big(\sqrt{\widehat{a}_{11}\widehat{a}_{22}} - \widehat{a}_{12}\big)}{\widehat{a}_{11} + \widehat{a}_{22}},$$

and multiplying numerator and denominator by $\sqrt{\widehat{a}_{11}\widehat{a}_{22}} + \widehat{a}_{12}$, then using $\widehat{a}_{11}\widehat{a}_{22} - \widehat{a}_{12}^2 = \det \widehat{A}_2$ and once more $\tau\widehat{a}_{11} = \sqrt{\widehat{a}_{11}\widehat{a}_{22}}$, produces (36). No sign is assumed for $\widehat{a}_{12}$, for $\det \widehat{A}_2$ or for ν_2; the hypotheses are $\widehat{a}_{11}, \widehat{a}_{22} > 0$ and, for the displayed quotient, $\sqrt{\widehat{a}_{11}\widehat{a}_{22}} + \widehat{a}_{12} \neq 0$.

The harmonic-mean bound, conditional. Assume $\widehat{A}_2 \succ 0$, so that $\nu_1 \ge \nu_2 > 0$. Substituting $\operatorname{tr} \widehat{A}_2 = \widehat{a}_{11} + \widehat{a}_{22} = \nu_1 + \nu_2$ and $\det \widehat{A}_2 = \nu_1\nu_2$ identifies the middle expression of (37) as the harmonic mean of the two eigenvalues, and

$$\frac{2\nu_1\nu_2}{\nu_1 + \nu_2} - \nu_2 = \frac{\nu_2(\nu_1 - \nu_2)}{\nu_1 + \nu_2} \;\ge\; 0, \qquad 2\nu_2 - \frac{2\nu_1\nu_2}{\nu_1 + \nu_2} = \frac{2\nu_2^2}{\nu_1 + \nu_2} \;\ge\; 0,$$

which are the two inequalities of (37). Both use $\nu_2 > 0$, and the hypothesis cannot be dropped: for $\widehat{A}_2 = \left[\begin{smallmatrix} 3 & 4 \\ 4 & 1 \end{smallmatrix}\right]$ one has $\operatorname{tr} \widehat{A}_2 = 4 > 0$ and $\det \widehat{A}_2 = -13$, so the middle expression of (37) equals $-13/2$ while $\nu_2 = 2 - \sqrt{17} = -2.123\ldots$; the claimed upper bound fails by a wide margin. That is why the hypothesis is named in the statement, and why Table 1 types the two halves of this proposition differently. ☐

Remark 5.11 (observation: the section is a closed loop)**.** Every statement of this section is an *identity* — with the single exception of the Gershgorin bound of Lemma 5.9, which is an inequality about ν_1 and not about ν_2, and of the second half of Proposition 5.10, which carries a named hypothesis. Reading them together:

- the constrained minimum at $K = 2$ is $G_{11}(1 - r_{12}^2)$ (Proposition 5.1), so the value of the optimal trial vector is the scalar $1 - r_{12}^2$ up to the closed factor G_{11};
- that scalar is the normalised determinant $\det \widehat{A}_2/(\widehat{a}_{11}\widehat{a}_{22})$ (Lemma 5.8), with no positivity used;
- the entrywise perturbation threshold is the same scalar up to the factor $1 + 3r_{12}^2 \in [1, 4]$ (Proposition 5.4);
- the only member of the Rayleigh family $u(t) = (t, -1)$ that meets the threshold, $t^\star = \sqrt{\widehat{a}_{22}/\widehat{a}_{11}}$, reintroduces $\det \widehat{A}_2$ exactly (Proposition 5.10);

- and the Wronskian and Lagrange identities (Lemmata 5.6–5.7) say that none of the smallness can be attributed to the two modes.

Consequently *no stage of the reduction lowers the accuracy demanded of* $\det \widehat{A}_2$: the four readings differ by the explicit factors G_{11}, $\widehat{a}_{11}\widehat{a}_{22}$, $1+3r_{12}^2$ and the closed prefactor of (36), all of which are computable from the three entries and none of which is small. We record this as an *observation* about a family of identities, not as a theorem: it is a reading, the statement "every exact reformulation costs the same precision" quantifies over reformulations rather than over windows, and how much precision is in fact demanded is a measured number that belongs in Section 7.

6 The exact bilinear form and the rank classification

This section writes $\widehat{A}_2$ out in terms of the comb and identifies the exact algebraic shape of the resulting object. Throughout the section the comb nodes are indexed by ℓ, so that i, j stay free for matrix indices.

Proposition 6.1 (exact split)**.** *Let D be admissible, $D < \min_\ell u_\ell$, and for a node u_ℓ let $m_\ell \in \mathbb{N}$ and $\theta_\ell \in [0,1)$ be defined by $u_\ell/D = m_\ell + \theta_\ell$. Put $Y^{(m)} = \Pi^\mathsf{T} T_M(\delta_m)\Pi$ with δ_m the lag sequence supported at the single lag m, and $Y^{(m)} := 0$ for $m \ge M$. Then, with $B_{ij} = t_i^\mathsf{T} A_{\mathrm{arch}} t_j$ and*

$$X_{ij}^{(\ell)} = \tfrac{1}{2}\Big[(1-\theta_\ell)\, t_i^\mathsf{T} Y^{(m_\ell)} t_j + \theta_\ell\, t_i^\mathsf{T} Y^{(m_\ell+1)} t_j\Big], \qquad S = \sum_\ell \varrho_\ell X^{(\ell)} \qquad (i,j \in \{1,2\}), \tag{38}$$

one has $\widehat{A}_2 = B - S$ exactly, and each of S_{11}, S_{22}, S_{12} is a linear *functional of the weight vector $(\varrho_\ell)_\ell$.*

Proof. Linearity. The three maps $c \mapsto T_M(c)$, $X \mapsto \Pi^\mathsf{T} X \Pi$ and $Y \mapsto (t_i^\mathsf{T} Y t_j)_{i,j\le 2}$ are linear, so the split (3) of the lag sequence induces the corresponding split of the 2×2 block. Writing δ_m for the lag sequence supported at the single lag m we have $c = c^{\mathrm{arch}} - \sum_\ell \varrho_\ell \varphi^{(\ell)}$ with $c_m^{\mathrm{arch}} = a(mD)$ and $\varphi_m^{(\ell)} = S_D(mD, u_\ell)$, and therefore $\widehat{A}_2 = B - S$ with $S_{ij} = \sum_\ell \varrho_\ell \sum_{m=0}^{M-1} \varphi_m^{(\ell)}\, t_i^\mathsf{T} Y^{(m)} t_j$.

The two-point read. Fix ℓ and $m \ge 0$. By definition $S_D(mD, u_\ell) = \frac{1}{2}\big[\Delta_D(mD - u_\ell) + \Delta_D(mD + u_\ell)\big]$, and the reflected hat vanishes identically: $mD + u_\ell \ge u_\ell > D$ by admissibility, while Δ_D is supported in $(-D, D)$. For the direct hat, $\Delta_D(mD - u_\ell) = \max\{0, 1 - |m - m_\ell - \theta_\ell|\}$, which is nonzero only when $|m - m_\ell - \theta_\ell| < 1$, i.e. only for $m \in \{m_\ell, m_\ell+1\}$, where it takes the values $1-\theta_\ell$ and θ_ℓ — including the case $\theta_\ell = 0$, where the second value is 0 and only one lag is read. Admissibility also gives $m_\ell \ge 1$, so the lag $m = 0$ is never touched by any node. The factor $\frac{1}{2}$ in (38) is the symmetrisation of S_D, and a lag index $\ge M$ falls outside the window and contributes nothing because $Y^{(m)} = 0$ there.

The lag-weight reading. By Lemma 2.9 applied to $v = t_i$, the coefficient of c_m in $t_i^\mathsf{T} A t_i$ is the lag weight $w_m(t_i)$ of (15); polarising in i and j,

$$t_i^\mathsf{T} Y^{(m)} t_j = w_m^{(ij)}, \qquad w^{(ij)} := \tfrac{1}{2}\big[w(t_i + t_j) - w(t_i) - w(t_j)\big], \tag{39}$$

so each entry of $X^{(\ell)}$ is the two-point read of *one closed weight vector* at the node position u_ℓ, and nothing else. This is the form used in Section 6.1.

Linearity in the weights. Each $S_{ij} = \sum_\ell \varrho_\ell X_{ij}^{(\ell)}$ is by construction a linear functional of $(\varrho_\ell)_\ell$, with coefficients $X_{ij}^{(\ell)}$ that depend on the window, the resolution and the node positions but not on the weights. □

Proposition 6.2 (polarisation of the determinant)**.** *With $D(P,Q) = P_{11}Q_{22} + P_{22}Q_{11} - 2P_{12}Q_{12}$,*

$$\det \widehat{A}_2 = \det B - D(B,S) + \det S, \tag{40}$$

where the three terms are respectively archimedean–archimedean, linear in the comb weights, and bilinear in them. Moreover

$$\det S = \sum_{\ell,\ell'} \varrho_\ell \varrho_{\ell'}\, \mathcal{K}(\ell,\ell'), \qquad \mathcal{K}(\ell,\ell') = \tfrac{1}{2} D(X^{(\ell)}, X^{(\ell')}), \tag{41}$$

with $X^{(\ell)}$ *the* 2×2 *matrix of* (38).

Proof. The polarisation. Expand directly:

$$\det(B-S) = (B_{11}-S_{11})(B_{22}-S_{22})-(B_{12}-S_{12})^2 = \det B-(B_{11}S_{22}+B_{22}S_{11}-2B_{12}S_{12})+\det S,$$

which is (40); it is also the instance $P = B$, $Q = S$ of Theorem 6.3(i) below, whose proof is independent of this proposition. By Proposition 6.1 the three terms are of degree 0, 1 and 2 in the weight vector respectively.

The double sum. D is symmetric and visibly linear in each argument, and $D(S, S) = 2S_{11}S_{22} - 2S_{12}^2 = 2 \det S$. Hence with $S = \sum_\ell \varrho_\ell X^{(\ell)}$,

$$\det S = \tfrac{1}{2} D(S, S) = \tfrac{1}{2} \sum_{\ell,\ell'} \varrho_\ell \varrho_{\ell'}\, D(X^{(\ell)}, X^{(\ell')}) = \sum_{\ell,\ell'} \varrho_\ell \varrho_{\ell'}\, \mathcal{K}(\ell,\ell'),$$

which is (41) [17, 2]. The kernel is symmetric, $\mathcal{K}(\ell,\ell') = \mathcal{K}(\ell',\ell)$, and its diagonal is $\mathcal{K}(\ell,\ell) = \frac{1}{2} D(X^{(\ell)}, X^{(\ell)}) = \det X^{(\ell)}$ — a fact used twice below, in Lemma 6.4 and in Measurement 6.8. □

Theorem 6.3 (rank three)**.** *Identify a symmetric* 2×2 *matrix* X *with its coordinate vector* $y(X) = (X_{11}, X_{22}, X_{12}) \in \mathbb{R}^3$ *and let*

$$J = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -2 \end{pmatrix}. \tag{42}$$

Then:

(i) $\det X = \frac{1}{2}\, y(X)^\mathsf{T} J y(X)$ *and* $D(P, Q) = y(P)^\mathsf{T} J y(Q)$*; in particular* D *is the polarisation of* det *and* $\det(P - Q) = \det P - D(P, Q) + \det Q$.

(ii) J *has eigenvalues* $\{1, -1, -2\}$*: it has* ***rank three*** *and* ***signature*** $(1, 2)$.

(iii) *Let* $P \in \mathbb{R}^{3\times n}$ *have columns* $y(X^{(\ell)})$, $\ell = 1, \ldots, n$*. Then the kernel of* (41) *is* $\mathcal{K} = \frac{1}{2} P^\mathsf{T} J P$*, so*

$$\operatorname{rank} \mathcal{K} \;\le\; 3 \qquad \textit{for every window, every } h \textit{ and every truncation,} \tag{43}$$

and the nonzero spectrum of $\mathcal{K}$ *equals the nonzero spectrum of the* 3×3 *matrix* $\frac{1}{2} J P P^\mathsf{T}$.

(iv) *Moreover* $\mathcal{K}$ *has at most* one *positive and at most* two *negative eigenvalues, for every* n *and every* P*: the signature of* J *is inherited, so the kernel of* (41) *is necessarily indefinite as soon as it has rank* 2 *or more.*

(v) *Consequently* $\det S = S_{11}S_{22} - S_{12}^2$*: the comb–comb term of* (40) *is a quadratic polynomial in exactly* three *linear functionals of the comb weights, and depends on the weight vector* (ϱ_ℓ) *through nothing else.*

Proof. (i) With $y = y(X)$, $\frac{1}{2} y^\mathsf{T} J y = \frac{1}{2}[2X_{11}X_{22} - 2X_{12}^2] = X_{11}X_{22} - X_{12}^2 = \det X$. Reading off the associated bilinear form,

$$y(P)^\mathsf{T} J y(Q) = P_{11}Q_{22} + P_{22}Q_{11} - 2P_{12}Q_{12} = D(P, Q),$$

so D is the polarisation of det, and since y is linear,

$$\det(P - Q) = \tfrac{1}{2}\big(y(P) - y(Q)\big)^\mathsf{T} J \big(y(P) - y(Q)\big) = \det P - y(P)^\mathsf{T} J y(Q) + \det Q,$$

which is the third relation.

(ii) J is block diagonal: the upper-left block $\left(\begin{smallmatrix}0&1\\1&0\end{smallmatrix}\right)$ has eigenvalues ± 1 with eigenvectors $(1,\pm 1,0)$, and the remaining diagonal entry is -2 with eigenvector $(0,0,1)$. Hence $\operatorname{spec} J = \{1,-1,-2\}$, all nonzero, so $\operatorname{rank} J = 3$, and the inertia is $(n_+,n_-) = (1,2)$.

(iii) By (i), $\mathcal{K}(\ell,\ell') = \frac{1}{2}D(X^{(\ell)},X^{(\ell')}) = \frac{1}{2}y(X^{(\ell)})^\top J y(X^{(\ell')})$, which is the (ℓ,ℓ') entry of $\frac{1}{2}P^\top JP$; and $\operatorname{rank}(P^\top JP) \le \operatorname{rank} J = 3$ whatever n and whatever the columns of P are. For the spectra, apply $\operatorname{spec}(UV)\setminus\{0\} = \operatorname{spec}(VU)\setminus\{0\}$ with $U = P^\top$ and $V = \frac{1}{2}JP$: then $UV = \mathcal{K}$ and $VU = \frac{1}{2}JPP^\top$. The latter is a product of the symmetric matrices J and $PP^\top \succeq 0$, so its spectrum is real, as it must be.

(iv) Suppose $\mathcal{K}$ had two positive eigenvalues. Then there is a two-dimensional $Z \subseteq \mathbb{R}^n$ with $z^\top \mathcal{K} z > 0$ for every $z \in Z\setminus\{0\}$. For such z this forces $Pz \neq 0$, so $P|_Z$ is injective and PZ is two-dimensional; and every $0 \neq \eta \in PZ$ is $\eta = Pz$ for some $0 \neq z \in Z$, so $\eta^\top J\eta = 2z^\top\mathcal{K}z > 0$. Thus J would be positive definite on a two-dimensional subspace, contradicting $n_+(J) = 1$ by Sylvester's law of inertia [22] in the Courant–Fischer form [5]. The same argument with three and $n_-(J) = 2$ gives the second half.

(v) Apply (i) to $X = S$: $\det S = S_{11}S_{22} - S_{12}^2$, and the three coordinates of $y(S)$ are the three linear functionals of Proposition 6.1. Since $\det S$ is a function of $y(S)$ alone, it depends on (ϱ_ℓ) through nothing else. □

Part (ii) will strike a reader as elementary — the determinant of a 2×2 matrix is visibly a quadratic form in its three entries — and it is. What the theorem is for is parts (iv)–(v) together with the corollary below, which is where the elementary observation has teeth. Before that, one reading of the kernel deserves to be isolated, because it is the source of the mixed signs.

Lemma 6.4 (the wedge form of the kernel, under exact rank one)**.** *Write $x\wedge x' = x_1x_2' - x_2x_1'$ for $x,x' \in \mathbb{R}^2$. If every read is of rank one, $X^{(\ell)} = \epsilon_\ell\, x_\ell x_\ell^\top$ with $\epsilon_\ell \in \{\pm 1\}$ and $x_\ell \in \mathbb{R}^2$, then*

$$\mathcal{K}(\ell,\ell') = \tfrac{1}{2}\,\epsilon_\ell\epsilon_{\ell'}\,\big(x_\ell \wedge x_{\ell'}\big)^2. \tag{44}$$

In particular the diagonal vanishes, $\mathcal{K}(\ell,\ell) = 0$, and the sign pattern of $\mathcal{K}$ is exactly that of $\epsilon_\ell\epsilon_{\ell'}$ — mixed as soon as the ϵ_ℓ are not all equal. The hypothesis is equivalent to $\det X^{(\ell)} = 0$ for every ℓ and is not *an identity: Measurement 6.8 reports how far from it the reads are.*

Proof. For $P = \epsilon aa^\top$ and $Q = \epsilon' bb^\top$,

$$D(P,Q) = \epsilon\epsilon'\big[a_1^2b_2^2 + a_2^2b_1^2 - 2a_1a_2b_1b_2\big] = \epsilon\epsilon'\big(a_1b_2 - a_2b_1\big)^2,$$

and $\mathcal{K} = \frac{1}{2}D$. The diagonal is $\frac{1}{2}(x_\ell\wedge x_\ell)^2 = 0$. That the hypothesis is equivalent to $\det X^{(\ell)} = 0$ is the classification of symmetric 2×2 matrices of rank at most one, and $\mathcal{K}(\ell,\ell) = \det X^{(\ell)}$ by the last line of the proof of Proposition 6.2, so the diagonal vanishes if and only if the hypothesis holds. □

Definition 6.5 (norm-factorised majorants)**.** A bound on the comb–comb term is *norm-factorised* if there are a norm $\mathcal{N}$ on $\mathbb{R}^n$ and a nondecreasing $\Phi : [0,\infty) \to [0,\infty)$ with

$$|\det S| \;\le\; \Phi(\mathcal{N}(\varrho)) \qquad \text{for every admissible weight vector } \varrho. \tag{45}$$

The class of all such bounds, over all norms and all nondecreasing Φ, is written $\mathfrak{M}$.

Corollary 6.6 (what a majorant of the comb–comb term can see)**.** (i) *If* (45) *holds then necessarily*

$$\Phi(t) \;\ge\; \sup\{|y_1y_2 - y_3^2| :\; y \in \mathbb{R}^3,\; y = (S_{11},S_{22},S_{12}) \text{ for some } \varrho \text{ with } \mathcal{N}(\varrho) \le t\}. \tag{46}$$

(ii) *Consequently no member of* $\mathfrak{M}$ *distinguishes two weight vectors with the same image* $y(S) = (S_{11}, S_{22}, S_{12})$*, and every cancellation between the three terms of* (40) *is invisible to it: the bound sees a point of* $\mathbb{R}^3$*, the phenomenon is a relation between that point and* B*.*

(iii) *If a route proceeds through a declared linear decomposition* $\varrho = \sum_{q \le p} \varrho^{(q)}$ *and bounds* $|\det S| \le \sum_{q,q'} \Phi_{qq'}(\mathcal{N}(\varrho^{(q)}), \mathcal{N}(\varrho^{(q')}))$ *with each* $\Phi_{qq'}$ *nondecreasing in both arguments, then each summand obeys the analogue of* (i) *for the corresponding block of the bilinear form, and the total pays in addition the triangle inequality across the* p^2 *pairs — whose cost is measured in Measurement 6.8.*

Proof. (i) By Theorem 6.3(v), $\det S = y_1 y_2 - y_3^2$ at $y = y(S)$. Fix $t > 0$ and let ϱ satisfy $\mathcal{N}(\varrho) \le t$. Then (45) and monotonicity of Φ give $|y_1 y_2 - y_3^2| \le \Phi(\mathcal{N}(\varrho)) \le \Phi(t)$; taking the supremum over all such ϱ gives (46).

(ii) $\varrho \mapsto y(S)$ is linear and, for $n \ge 4$, has a kernel of dimension at least $n - 3$; two weight vectors differing by an element of that kernel have the same $\det S$ and the same $\det \widehat{A}_2$, while (45) sees them only through $\mathcal{N}$. That the three terms of (40) may cancel to any order without Φ noticing is then immediate, since Φ does not involve B at all.

(iii) Apply (i) to each pair (q, q') with S replaced by the pair of blocks, using bilinearity of D in (41); the last clause is the statement that $|\sum_{q,q'} \cdot| \le \sum_{q,q'} |\cdot|$ has been used, whose slack is the ℓ^1-versus-signed ratio measured below. □

Remark 6.7 (the five declared routes, classified)**.** Of the five preregistered routes of Measurement 7.6, three are members of $\mathfrak{M}$ as they are stated: the large-sieve/duality route, with $\mathcal{N} = \|\cdot\|_2$ and $\Phi(t) = \|\mathcal{K}\|_{\mathrm{op}} t^2$; the Montgomery–Vaughan route, which bounds each of the three linear functionals through $\|\varrho\|_2$ — its interpolation residual through $\|\varrho\|_1 \le \sqrt{n}\,\|\varrho\|_2$ on a window with n nodes, so that one norm suffices — and then combines them by $|\det S| \le |S_{11}||S_{22}| + S_{12}^2$; and the comb–comb term of the exact split $|\det B| + |D(B,S)| + |\det S|$. Vaughan's Type I + Type II route is of the shape of Corollary 6.6(iii) with $p = 4$. The remaining route, Cauchy–Schwarz on $\widehat{A}_2$, is not a majorant of $\det S$ at all: it asserts $\widehat{a}_{12}^2 \le \widehat{a}_{11}\widehat{a}_{22}$, i.e. $\det \widehat{A}_2 \ge 0$, and is carried in the table as the trivial baseline. This classification is a *reading of the five routes as declared*, not a theorem about every conceivable implementation of them; what is a theorem is Corollary 6.6 itself.

Measurement 6.8 (mixed signs, and what the triangle inequality costs)**.** Two numbers price the two hypotheses just used. *The rank-one defect*: the median of $|\det X^{(\ell)}|/\|X^{(\ell)}\|^2$ over the nodes is $1.3 \cdot 10^{-4}$ on the reference window of Measurement 6.16, and on Surface I it lies between $4.6 \cdot 10^{-3}$ and $3.1 \cdot 10^{-2}$, below the declared bar $5 \cdot 10^{-2}$ on every window. So the wedge reading of Lemma 6.4 is accurate but not exact — the diagonal of $\mathcal{K}$ is small, not zero, and it carries about 6% of $\det S$ on the reference window. *The* ℓ^1 *mass*: on a declared node subsample of the reference window,

$$\sum_{\ell,\ell'} \varrho_\ell \varrho_{\ell'} |\mathcal{K}(\ell,\ell')| \;=\; 40.6 \times \Big|\sum_{\ell,\ell'} \varrho_\ell \varrho_{\ell'} \mathcal{K}(\ell,\ell')\Big|, \tag{47}$$

the subsample being the 900 nodes of the declared horizon and the ratio being reported on the window it was computed on and on no other. A route that replaces the signed double sum by its absolute value therefore loses that factor before it begins, which is the measured content of Corollary 6.6(iii); the mixed signs it loses are the ones Lemma 6.4 and Theorem 6.3(iv) predict.

6.1 The bandwidth of the weight kernel

By Proposition 6.1 an entry of $X^{(\ell)}$ is the two-point read of one closed weight vector at the single real argument $\tau_\ell = u_\ell / D$. This subsection identifies that weight vector in closed form, extracts the consequence for a factorisation-indexed block, and says exactly where the argument stops. Throughout, τ denotes a real lag argument, $N = 2h + 1$ and $\omega_k = 2\pi k/N$ as in (5).

Lemma 6.9 (the closed lag weights of a parity mode). *For $1 \le k \le h$ the lag weights (15) of the parity mode t_k are $w_0(t_k) = 1$ and*

$$w_d(t_k) = \frac{2}{N}\Big[(N-d)\cos(\omega_k d) + \cot(\omega_k)\,\sin(\omega_k d)\Big], \qquad 1 \le d \le M-1, \tag{48}$$

one formula for the whole range, with no change of branch at $d = h$.

Proof. Apply Lemma 2.9 with $c = \delta_d$ the lag sequence supported at the single lag d, and let $A_{(d)}$ be the odd window form (2) of δ_d: since $\sum_e c_e w_e = w_d$ for this c, we get $w_d(v) = v^\mathsf{T} A_{(d)} v$, i.e. by (2)

$$w_d(v) = \sum_{r,r'=0}^{h-1} v_r v_{r'}\Big(\mathbf{1}[\,|r-r'| = d\,] - \mathbf{1}[\,r + r' = M-1-d\,]\Big). \tag{49}$$

Put $T_x = \sin(\omega_k x)$ for $x \in \mathbb{Z}$. Then T is N-periodic and odd, so $T_{N-m} = -T_m$, and $(t_k)_r = (2/\sqrt{N})T_{r+1}$. Substituting $x = r+1$, $y = r'+1 \in [1,h]$ in (49), the Hankel condition $r + r' = M-1-d$ becomes $x + y = N - d$, where $T_y = T_{N-d-x} = -T_{x+d}$, so the Hankel term enters with a + sign; the constraint $y \in [1,h]$ becomes $x \in [h+1-d, 2h-d]$. The Toeplitz term contributes $\sum_{|x-y|=d} T_x T_y = 2\sum_{x=1}^{h-d} T_x T_{x+d}$, each pair being counted twice for $d \ge 1$. Hence, for $d \ge 1$,

$$w_d(t_k) = \frac{4}{N}\Big[2\sum_{x=1}^{h-d} T_x T_{x+d} + \sum_{x \in I_d} T_x T_{x+d}\Big], \qquad I_d = [h+1-d,\, 2h-d] \cap [1,h]. \tag{50}$$

Write $\Phi(m) = \sum_{x=1}^{m} T_x T_{x+d}$. For $1 \le d \le h$ one has $I_d = [h+1-d, h]$, so the bracket is $2\Phi(h-d) + [\Phi(h) - \Phi(h-d)] = \Phi(h-d) + \Phi(h)$; for $h < d \le M-1$ the first sum is empty and $I_d = [1, 2h-d]$, so the bracket is $\Phi(2h-d)$.

For Φ, use $2\sin a \sin b = \cos(a-b) - \cos(a+b)$ and the Dirichlet sum $\sum_{x=1}^{m}\cos(a + 2\omega x) = \frac{\sin(m\omega)}{\sin\omega}\cos(a + (m+1)\omega)$, legitimate because N is odd and $1 \le k \le h$, so $\sin\omega_k \ne 0$ and $2\omega_k \notin 2\pi\mathbb{Z}$:

$$\Phi(m) = \frac{m}{2}\cos(\omega d) - \frac{1}{2}\,\frac{\sin(m\omega)}{\sin\omega}\,\cos\big(\omega(m+d+1)\big), \qquad \omega = \omega_k. \tag{51}$$

Now evaluate, using $\omega N = 2\pi k$ and $h = (N-1)/2$, so that $\omega h = \pi k - \omega/2$. At $m = h$: $\sin(\omega h) = -(-1)^k\sin(\omega/2)$ and $\cos(\omega(h+d+1)) = (-1)^k\cos(\omega(d+\frac{1}{2}))$, whence $\Phi(h) = \frac{N}{4}\cos(\omega d) - \frac{1}{4}\tan(\frac{\omega}{2})\sin(\omega d)$ after expanding $\cos(\omega(d+\frac{1}{2}))$. At $m = h-d$: $\sin(\omega(h-d)) = -(-1)^k\sin(\omega(d+\frac{1}{2}))$ and $\cos(\omega(h+1)) = (-1)^k\cos(\omega/2)$, whence $\Phi(h-d) = \frac{N-2d}{4}\cos(\omega d) + \frac{1}{4}\cot(\frac{\omega}{2})\sin(\omega d)$. Adding, and using $\cot(\frac{\omega}{2}) - \tan(\frac{\omega}{2}) = 2\cot\omega$,

$$\Phi(h-d) + \Phi(h) = \frac{N-d}{2}\cos(\omega d) + \frac{1}{2}\cot(\omega)\sin(\omega d),$$

which with the factor $4/N$ is (48). At $m = 2h-d = N-1-d$ one has $\sin(\omega m) = -\sin(\omega(d+1))$ and $\cos(\omega(m+d+1)) = 1$, so $\Phi(2h-d) = \frac{N-d}{2}\cos(\omega d) + \frac{1}{2}\cot(\omega)\sin(\omega d)$ as well — the same expression, which is why (48) has no branch at $d = h$. Finally $w_0 = \|t_k\|^2 = 1$ by (16); the right-hand side of (48) equals 2 at $d = 0$, the discrepancy being the doubling of the lag-d pairs, which does not occur at $d = 0$. $\square$

Lemma 6.10 (the weight sequence has bandwidth K and degree one). *Let $\mathcal{V}_K \subset \mathbb{R}^{M-1}$ be the span of the $4K$ sequences*

$$d \mapsto \cos(\omega_k d), \quad \sin(\omega_k d), \quad d\cos(\omega_k d), \quad d\sin(\omega_k d), \qquad k = 1,\dots,K,\ 1 \le d \le M-1. \tag{52}$$

Then $(w_d(v))_{d=1}^{M-1} \in \mathcal{V}_K$ *for every* $v \in \operatorname{span}\{t_1, \dots, t_K\}$*, and consequently also for every polarised weight vector* $w^{(ij)}$ *of* (39) *with* $i, j \le K$*. The coefficients of* $d\cos(\omega_k d)$ *and* $d\sin(\omega_k d)$ *are proportional to* a_k^2*, where* $v = \sum_k a_k t_k$*.*

Proof. Repeat the proof of Lemma 6.9 with T_x replaced by $V_x = \sum_{k \le K} a_k \sin(\omega_k x)$: (50) holds verbatim with $\Phi_v(m) = \sum_{x=1}^m V_x V_{x+d} = \sum_{k,l} a_k a_l \sum_{x=1}^m \sin(\omega_k x)\sin(\omega_l(x+d))$. For $k = l$ the inner sum is (51), contributing the four sequences of (52) at frequency ω_k, the linear-in-d ones with coefficient proportional to a_k^2 through the term $\frac{m}{2}\cos(\omega d)$ at $m = h - d$ resp. $m = 2h - d$. For $k \ne l$ write $2\sin(\omega_k x)\sin(\omega_l x + \omega_l d) = \cos(\lambda_- x - \omega_l d) - \cos(\lambda_+ x + \omega_l d)$ with $\lambda_\pm = \omega_k \pm \omega_l$, and sum by $\sum_{x=1}^m \cos(a + \lambda x) = \frac{\sin(\lambda m/2)}{\sin(\lambda/2)}\cos(a + \lambda(m+1)/2)$, legitimate since $0 < |k \pm l| < N$ gives $\sin(\lambda_\pm/2) \ne 0$. Each such term is a product $\sin P \cos Q = \frac{1}{2}[\sin(P+Q) + \sin(P-Q)]$ in which P and Q are affine in d — the evaluation points $m \in \{h, h-d, 2h-d\}$ are affine in d with slope 0 or -1 — and a direct computation of the slopes, using $\omega_k N, \omega_l N \in 2\pi\mathbb{Z}$, gives $\operatorname{slope}(P+Q) = -\omega_k$ and $\operatorname{slope}(P-Q) = +\omega_l$ in both cases $\lambda_\pm$ and at all three evaluation points. So the off-diagonal terms contribute only the frequencies ω_k, ω_l with constant coefficients, and no linear-in-d factor appears because $\sin(\lambda m/2)/\sin(\lambda/2)$ is bounded, not linear. The statement for $w^{(ij)}$ follows because $\mathcal{V}_K$ is a linear space and $w^{(ij)} = \frac{1}{2}[w(t_i + t_j) - w(t_i) - w(t_j)]$ with all three arguments in $\operatorname{span}\{t_1, \dots, t_K\}$. □

The next lemma is the general mechanism, stated once and independently of the weights. It is the statement asked for in the source material: bandwidth in an additive variable implies rank at a declared truncation error.

Lemma 6.11 (bandwidth ⇒ η-rank)**.** *Let* $x_1, \dots, x_R$ *and* $y_1, \dots, y_C$ *be real and let* $I \subseteq \mathbb{R}$ *contain every* $x_b + y_e$*. Let* $\Omega \ge 0$ *be the number of frequencies* $\nu_1, \dots, \nu_\Omega > 0$*, let* $q \ge 0$ *be a degree bound, and let*

$$W_\Omega(\tau) = p_0(\tau) + \sum_{j=1}^{\Omega}\big[p_j(\tau)\cos(\nu_j\tau) + \tilde{p}_j(\tau)\sin(\nu_j\tau)\big], \qquad \deg p_j, \deg \tilde{p}_j \le q. \tag{53}$$

If $W : I \to \mathbb{R}$ *satisfies* $\sup_{\tau\in I}|W(\tau) - W_\Omega(\tau)| \le \eta$*, then the matrix* $F_{be} = W(x_b + y_e)$ *obeys*

$$\sigma_{r+1}(F) \;\le\; \eta\sqrt{RC}, \qquad r = (2\Omega+1)(q+1); \tag{54}$$

if the expansion (53) *has no polynomial term* p_0*, then* $r = 2\Omega(q+1)$ *suffices. Row and column rescalings* $F \mapsto \operatorname{diag}(\alpha)F\operatorname{diag}(\beta)$ *leave* r *unchanged and replace the right-hand side by* $\eta\|\alpha\|_2\|\beta\|_2$*.*

Proof. Let F_Ω be the matrix built from W_Ω. Fix j and $a \le q$. Since $(x+y)^a = \sum_{p\le a}\binom{a}{p}x^p y^{a-p}$ and $e^{i\nu(x+y)} = e^{i\nu x}e^{i\nu y}$, the function $\tau^a e^{i\nu_j\tau}$ of $\tau = x + y$ is a finite sum of products $[x^p e^{i\nu_j x}] \cdot [y^{a-p}e^{i\nu_j y}]$. Taking real and imaginary parts, the columns of the matrices $\big[\tau^a\cos(\nu_j\tau)\big]_{be}$ and $\big[\tau^a\sin(\nu_j\tau)\big]_{be}$ lie in the span of the $2(q+1)$ vectors $\big(x_b^p\cos(\nu_j x_b)\big)_b$, $\big(x_b^p\sin(\nu_j x_b)\big)_b$, $p \le q$. Likewise the columns of $[p_0(x_b + y_e)]_{be}$ lie in the span of the $q+1$ vectors $(x_b^p)_b$. Summing over j, $\operatorname{rank} F_\Omega \le 2\Omega(q+1) + (q+1) = (2\Omega+1)(q+1)$, and the p_0-free case drops the last summand. Write $E = F - F_\Omega$; then $\|E\|_2 \le \|E\|_F \le \eta\sqrt{RC}$, and Weyl's perturbation inequality for singular values [25] gives $\sigma_{r+1}(F) \le \sigma_{r+1}(F_\Omega) + \|E\|_2 = \|E\|_2$ because $\sigma_{r+1}(F_\Omega) = 0$. For the rescaling, ranks are unchanged by invertible diagonal factors, and $|\alpha_b E_{be}\beta_e| \le \eta|\alpha_b||\beta_e|$ gives $\|\operatorname{diag}(\alpha)E\operatorname{diag}(\beta)\|_F \le \eta\|\alpha\|_2\|\beta\|_2$. □

What Lemma 6.11 needs is a band expansion of the *read*, and the read of Proposition 6.1 is not the sequence (w_d) but its piecewise-linear interpolant. That interpolation is the only source of η.

Lemma 6.12 (the two-point read against the analytic read)**.** *Let* $v \in \operatorname{span}\{t_1, \ldots, t_K\}$ *and let* W_v *be the function obtained by evaluating the representation of Lemma 6.10 at real argument, so that* $W_v(d) = w_d(v)$ *for integers* $1 \le d \le M-1$*. For* $1 \le \tau \le M-1$ *the two-point read* $\widehat{W}_v(\tau) = (1-\theta)w_m(v) + \theta w_{m+1}(v)$, $m = \lfloor \tau \rfloor$, $\theta = \tau - m$, *of* (38) *satisfies*

$$\sup_{1\le\tau\le M-1} |\widehat{W}_v(\tau) - W_v(\tau)| \;\le\; \tfrac{1}{8} \sup_{1\le\tau\le M-1} |W_v''(\tau)| \;\le\; \tfrac{1}{8}\big[\omega_K^2\Lambda_0 + 2\omega_K\Lambda_1\big], \tag{55}$$

where $\Lambda_0 = \sum_k(|\alpha_k| + |\gamma_k| + M(|\beta_k| + |\delta_k|))$ *and* $\Lambda_1 = \sum_k(|\beta_k| + |\delta_k|)$ *in the notation* $W_v(\tau) = \sum_k[(\alpha_k + \beta_k\tau)\cos(\omega_k\tau) + (\gamma_k + \delta_k\tau)\sin(\omega_k\tau)]$*. For the single mode* t_k*,*

$$\sup_{1\le\tau\le M-1} |\widehat{W}_{t_k}(\tau) - W_{t_k}(\tau)| \;\le\; \frac{\pi^2k^2}{N^2}\Big(1 + \frac{3}{2\pi k}\Big) \;=\; O\bigg(\frac{k^2}{N^2}\bigg). \tag{56}$$

Proof. The first inequality is the standard error of linear interpolation on a unit interval: for $f \in C^2[m, m+1]$ the error at $\tau = m+\theta$ is $-\frac{1}{2}\theta(1-\theta)f''(\xi)$ for some ξ, and $\max_\theta \theta(1-\theta) = \frac{1}{4}$. For the second, differentiate the representation twice: each term contributes $-\omega_k^2(\alpha_k+\beta_k\tau)\cos(\omega_k\tau) + 2\omega_k\beta_k(-\sin(\omega_k\tau))$ and its sine analogue, so $|W_v''| \le \omega_K^2\Lambda_0 + 2\omega_K\Lambda_1$ on $\tau \le M$. For a single mode, (48) gives $\alpha_k = 2$, $\beta_k = -2/N$, $\gamma_k = (2/N)\cot\omega_k$, $\delta_k = 0$, hence $|W_{t_k}''| \le \frac{2}{N}[\omega_k^2 N + 2\omega_k + \omega_k^2\cot\omega_k] \le 2\omega_k^2 + 6\omega_k/N$ using $\omega\cot\omega \le 1$; dividing by 8 and inserting $\omega_k = 2\pi k/N$ gives (56). □

Proposition 6.13 (η-rank of a factorisation-indexed block)**.** *Fix* K*, let the weight vector be either* $w(v)$ *for some* $v \in \operatorname{span}\{t_1, \ldots, t_K\}$ *or a polarised* $w^{(ij)}$ *with* $i, j \le K$*, and let* $\widehat{W}$ *be its two-point read as in Lemma 6.12. Let* $\mathcal{B}_1, \mathcal{B}_2$ *be two finite sets of reals* > 1*, put* $x_{b_1} = \log b_1/D$*,* $y_{b_2} = \log b_2/D$*, and assume the block is* interior*:*

$$1 \le x_{b_1} + y_{b_2} \le M-1 \qquad \text{for all } b_1 \in \mathcal{B}_1,\ b_2 \in \mathcal{B}_2. \tag{57}$$

Then the matrix of reads $F_{b_1b_2} = \widehat{W}(x_{b_1} + y_{b_2})$*,* $R = |\mathcal{B}_1|$*,* $C = |\mathcal{B}_2|$*, satisfies*

$$\sigma_{4K+1}(F) \;\le\; \eta\sqrt{RC}, \qquad \eta = \tfrac{1}{8}\sup_{1\le\tau\le M-1}|W''|, \tag{58}$$

with η *bounded by* (55)*; for a single mode* $\eta \le \pi^2k^2N^{-2}(1 + 3/(2\pi k))$*. The same bound holds for* $\operatorname{diag}(\alpha)F\operatorname{diag}(\beta)$ *with* η *replaced by* $\eta\|\alpha\|_2\|\beta\|_2$*, so arithmetic coefficients attached to the two index sets do not change the rank count.*

Proof. By Lemma 6.10 the analytic read W lies in the span of the $4K$ functions (52), which is an expansion (53) with $\Omega = K$ frequencies, degree bound $q = 1$ and no polynomial term p_0; by Lemma 6.12 $\sup|\widehat{W} - W| \le \eta$ on $[1, M-1]$, which by (57) contains every $x_{b_1} + y_{b_2}$. Lemma 6.11 with $r = 2\Omega(q+1) = 4K$ gives (58), and its last clause gives the rescaled version. □

Remark 6.14 (what (57) costs, and what the proposition does not say)**.** Hypothesis (57) is not cosmetic. Outside the window the weights are set to zero ($w_m := 0$ for $m \ge M$ in Proposition 6.1), and the truncation $\mathbf{1}[x + y \le M-1]$ is a triangular cut-off, which is *not* of low rank: without (57) the conclusion is false as stated, and what survives is the same bound for the interior sub-block plus the cut-off part. On the declared surfaces the node cap makes the block interior by construction, and that is how the hypothesis is met. Two further limits are worth naming. The bound (58) is a statement about $4K$ singular values with a declared η; it is not a statement that the block "has rank $O(1)$" in any asymptotic sense, and the $\sqrt{RC}$ factor means that the useful regime is $\eta\sqrt{RC} \ll \sigma_1(F)$, which is a condition on the declared surface and is checked there, not here. And the whole mechanism is about *additivity* of the lag argument under factorisation: it uses that the read happens at $\tau = u/D$ with $u = \log n$, so that $n = b_1b_2$ gives $\tau = x_{b_1} + y_{b_2}$. Nothing arithmetic beyond that additivity is used, and nothing is claimed about any Type II sum beyond the singular values of its weight matrix.

Measurement 6.15 (Type II blocks: the measured collapse inside the proved bound)**.** Proposition 6.13 bounds σ_{4K+1}; the surfaces report a much sharper collapse, and the two are typed separately. On the declared surface the Type II blocks of Vaughan's identity [24] built from the closed weights, with $K = 2$ so that the proved bound is $\sigma_9 \leq \eta\sqrt{RC}$ with $\eta \leq 2.6 \cdot 10^{-4}$ at $h = 301$ for each of the three polarised reads by (55), are measured to have 2 singular values carrying 99.9% of the mass, against 153 for a generic kernel of the same size, over the scanned range of the decomposition parameter; the four-term identity for Λ is verified coefficientwise first. The number 2 is measured and stays measured: the theorem gives 8 at $K = 2$, the observed sufficiency of 2 is a property of the declared surface and is not proved here.

Measurement 6.16 (the reference window: a three-term cancellation)**.** On the reference window $h = 540$, with the comb truncated at the node cap $2.465 \cdot 10^4$, the three terms of (40) are

$$\det B = 6.18, \qquad -D(B,S) = -198.8, \qquad \det S = 192.6, \qquad \text{sum} = 4.22 \cdot 10^{-3}, \tag{59}$$

i.e. five orders of magnitude of cancellation between three pieces of size ≈ 200; the absolute value demanded of $\det \widehat{A}_2$ at that window — the target against which the cancellation has to be resolved — is $1.10 \cdot 10^{-5}$, a further factor 384 below the residual sum. The four numbers are read under the per-window near-null convention (F1), and the caps are those of Surface II (Section 8.2): the window index inside $[142, 1445]$, the conditioning of the raw Gram block inside $\operatorname{cond}(G) \leq 10^{12}$, and the comb truncated at the node cap quoted above. Three readings of the same window follow from the results of this section and are recorded as part of the measurement: the cancellation is between the three terms of the exact polarisation (40), so no route of Definition 6.5 sees it (Corollary 6.6(ii)); the signed kernel sum behind $\det S$ is 40.6 times smaller than its ℓ^1 mass on the declared subsample (Measurement 6.8); and the sum in (59) is $\det \widehat{A}_2$, whose normalisation by $\widehat{a}_{11}\widehat{a}_{22}$ is the quantity of Problem 7.1 — so five orders of magnitude here is a statement about one window and about nothing else.

7 A measured obstruction, and an open problem

Sections 3–6 reduce the question of Section 1.1 to a single scalar and classify what the standard unconditional toolbox does to it. This section states the scalar question that remains, reports what is measured about it on declared surfaces, prices the routes, and locates — by controlled experiment — where the phenomenon actually lives. Everything here is either a proved statement about a route or a number on a named surface; the two are never mixed, and Table 1 is the binding record.

7.1 The open problem

Problem 7.1 (three equivalent forms)**.** Is there, for each $\varepsilon > 0$, a constant C_ε such that *uniformly in the window index,*

$$1 - r_{12}^2 = \frac{\det \widehat{A}_2}{\widehat{a}_{11}\widehat{a}_{22}} \;\leq\; C_\varepsilon\, h^{-3+\varepsilon}\,? \tag{60}$$

Equivalently, by Lemmata 5.8, 5.7 and 7.2:

(a) *Row collinearity.* Do the two rows of $\widehat{A}_2$ — two explicit finite sums against the closed weights of Proposition 6.1 — become collinear at rate $h^{-3+\varepsilon}$, the sine of the angle between them being $|\det \widehat{A}_2|/(\|\mathrm{row}_1\|\|\mathrm{row}_2\|)$?

(b) *Determinant form.* Does the normalised determinant $\det \widehat{A}_2/(\widehat{a}_{11}\widehat{a}_{22})$ satisfy (60)?

(c) *Joint near-degeneracy.* Does the triple (S_{11}, S_{22}, S_{12}) of Theorem 6.3(v) approach the archimedean block B jointly, at relative precision sharpening like h^{-3}, in the sense that the three terms of (40) cancel to that order?

The problem is open. It is a question about a finite bilinear form — for each window a polynomial identity in three explicit finite sums — and the only thing missing is a bound uniform in h. It is not formulated as, and must not be read as, an approach to, a weakening of, or a consequence of any conjecture; no comparison of strengths against any conjecture is made anywhere in this paper, and no such comparison is needed to state the problem or to read the measurements below.

The three forms are equivalent because the quantities in (a) and (b) differ by a factor that is bounded above and below in closed form. We record this once, because it is also what makes the two fitted exponents of Measurement 7.3 non-independent (Remark 8.2).

Lemma 7.2 (the angle reading against the determinant reading)**.** *Assume $\widehat{a}_{11}, \widehat{a}_{22} > 0$ and $|\widehat{a}_{12}| \le \sqrt{\widehat{a}_{11}\widehat{a}_{22}}$, and write $\rho_1 = \|(\widehat{a}_{11}, \widehat{a}_{12})\|$, $\rho_2 = \|(\widehat{a}_{12}, \widehat{a}_{22})\|$ for the two row norms of $\widehat{A}_2$ and $\sin\angle = |\det \widehat{A}_2|/(\rho_1\rho_2)$ for the sine of the angle between the rows. Then*

$$\frac{\sin\angle}{|1 - r_{12}^2|} = \frac{\widehat{a}_{11}\widehat{a}_{22}}{\rho_1\rho_2} = \frac{\sqrt{\widehat{a}_{11}\widehat{a}_{22}}}{\sqrt{(\widehat{a}_{11} + r_{12}^2\widehat{a}_{22})(\widehat{a}_{22} + r_{12}^2\widehat{a}_{11})}}, \tag{61}$$

and this factor obeys

$$\frac{1}{2}\cdot\frac{2\sqrt{\widehat{a}_{11}\widehat{a}_{22}}}{\widehat{a}_{11} + \widehat{a}_{22}} \le \frac{\sin\angle}{|1 - r_{12}^2|} \le 1, \tag{62}$$

so the two readings differ by a factor confined to $(0, 1]$ whose two-sided bound depends on the window only through the diagonal ratio $\widehat{a}_{22}/\widehat{a}_{11}$.

Proof. By Lemma 5.8, $|1 - r_{12}^2| = |\det \widehat{A}_2|/(\widehat{a}_{11}\widehat{a}_{22})$, and $\sin\angle = |\det \widehat{A}_2|/(\rho_1\rho_2)$ by definition; dividing gives the first equality of (61). For the second, substitute $\widehat{a}_{12}^2 = r_{12}^2\widehat{a}_{11}\widehat{a}_{22}$:

$$\rho_1^2 = \widehat{a}_{11}^2 + \widehat{a}_{12}^2 = \widehat{a}_{11}(\widehat{a}_{11} + r_{12}^2\widehat{a}_{22}), \qquad \rho_2^2 = \widehat{a}_{12}^2 + \widehat{a}_{22}^2 = \widehat{a}_{22}(\widehat{a}_{22} + r_{12}^2\widehat{a}_{11}),$$

and take the product. For (62): the upper bound is $\rho_1\rho_2 \ge \widehat{a}_{11}\widehat{a}_{22}$, which follows from $\rho_1 \ge \widehat{a}_{11}$ and $\rho_2 \ge \widehat{a}_{22}$; the lower bound uses $r_{12}^2 \le 1$, whence $\widehat{a}_{11} + r_{12}^2\widehat{a}_{22} \le \widehat{a}_{11} + \widehat{a}_{22}$ and likewise for the second factor, so $\rho_1\rho_2 \le \sqrt{\widehat{a}_{11}\widehat{a}_{22}}\,(\widehat{a}_{11} + \widehat{a}_{22})$. The final form is $\frac{1}{2}$ times the ratio of the geometric to the arithmetic mean of $\widehat{a}_{11}$ and $\widehat{a}_{22}$, i.e. $\sqrt{t}/(1 + t)$ with $t = \widehat{a}_{22}/\widehat{a}_{11}$. □

7.2 What is measured

Measurement 7.3 (measured rates)**.** All numbers here are least-squares fits of log of the quantity against $\log h$ on the stated finite surface, reported for orientation and used as bounds nowhere. On Surface II of Section 8.2 the readings are:

- $1 - r_{12}^2 \sim h^{-2.747}$ on the 67 node-complete windows, $h = 142$ to 1433;
- $1 - r_{12}^2 \sim h^{-2.260}$ on the full family of 70 windows, $h = 142$ to 1445, i.e. including the 3 windows whose comb is truncated by the node cap. Both numbers are quoted; the difference is the cap, not the phenomenon, and it is not hidden;
- the angle reading of Problem 7.1(a), $\sin\angle = |\det \widehat{A}_2|/(\rho_1\rho_2)$, falls as $h^{-2.711}$ on the same 67 windows;
- on the 27-window frame ladder of Measurements 3.6 and 4.5 the fitted exponent is -2.921; on the union of that ladder with the decoupled families below it is -2.246.

The exponent is *not* a single number across surfaces, and this is reported without smoothing. On three independently generated frames obtained by decoupling the cell width from the window index the fitted exponents are -1.702, -1.270 and -1.735, while the best priced route of Measurement 7.6 certifies $+0.529$, $+0.339$ and $+0.611$ there — short of the exponent in (60) by 1.17, 0.93 and 1.12. On five further frame data the pooled exponents are -2.19, -2.46, -2.55, -1.83 and -2.76, and across the nine individual ladders of those families they range from -1.83 to -2.87. What does transfer is the near-degeneracy itself: the row angle stays below 10^{-2} on every window of every family and decays on every ladder. The rate is a property of the family; the collapse is not. Nothing in Sections 2–6 depends on any of these numbers.

7.3 The priced routes, and the structural obstructions

The next proposition collects five unconditional majorants for $|1 - r_{12}^2|$. Each is an inequality with its hypotheses written out; each is proved; none of them involves a fit. The fitted exponents that these majorants *certify* on a declared surface are a separate matter and are reported afterwards, in Measurement 7.6.

One hypothesis of one of the five is not an assumption at all but a consequence of the assembly, and it is separated out first, because a sieve route has to be told which points it is applied at and the window decides them. The reads of Proposition 6.1 are values of one exponential sum at two points fixed by the parity modes and the resolution, and their spacing is closed.

Lemma 7.4 (the three linear functionals are dual-point reads)**.** *Let D be admissible, let $K = 2$, and take the prime-power instance of Section 2.1: $u_\ell = \log n_\ell$, $\varrho_\ell = 2\Lambda(n_\ell)/\sqrt{n_\ell}$ over the prime powers $n_\ell \le X$, with X the smaller of $e^{2\alpha}$ and the declared node cap, and with every read interior, $1 \le \tau_\ell = u_\ell/D \le M - 1$. Write the band representation of the polarised weight $w^{(ij)}$ of* (39)*, which exists by Lemma 6.10, as*

$$W^{(ij)}(\tau) = \sum_{k=1,2} \big[(a_k + b_k\tau)\cos(\omega_k\tau) + (c_k + d_k\tau)\sin(\omega_k\tau)\big], \tag{63}$$

its coefficients being those of Lemma 6.12 with the letters chosen so as to leave γ for the dual points, and let $\Lambda_0^{(ij)} = \sum_k(|a_k| + |c_k|) + M\sum_k(|b_k| + |d_k|)$ and $\eta^{(ij)}$ be the coefficient budget and the interpolation bound of (55) *for that weight. Put*

$$\gamma_k = \frac{\omega_k}{D} = \frac{2\pi k}{ND} \quad (k = 1, 2), \qquad \Delta\gamma = \gamma_2 - \gamma_1 = \frac{2\pi}{ND}, \qquad T_q(\gamma) = \sum_{n \le X} \varrho_n (\log n)^q n^{\mathrm{i}\gamma}. \tag{64}$$

Then

$$S_{ij} = \sum_{k=1,2} \mathrm{Re}\Big[(a_k - \mathrm{i}c_k)T_0(\gamma_k) + \tfrac{1}{D}(b_k - \mathrm{i}d_k)T_1(\gamma_k)\Big] + E_{ij}, \qquad |E_{ij}| \le \eta^{(ij)}\|\varrho\|_1, \tag{65}$$

so each of the three reads is a value of one exponential sum of length X, and of its first logarithmic moment, at the two points (64) *and at no others. If in addition the Montgomery–Vaughan inequality [18] is applied at those two points, in the form $\sum_{k\le 2}|T_0(\gamma_k)|^2 \le (X + \Delta\gamma^{-1})\|\varrho\|_2^2$ and the same with the weights $\varrho_n \log n$ for T_1, then*

$$|S_{ij}| \;\le\; \Lambda_0^{(ij)}\, Q + \eta^{(ij)}\|\varrho\|_1, \qquad Q^2 = (X + \Delta\gamma^{-1})\|\varrho\|_2^2. \tag{66}$$

Proof. By Lemma 6.10 at $K = 2$ the analytic read $W^{(ij)}$ lies in the span of the eight sequences (52) with $k \le 2$, which is (63). The proof of Lemma 6.12 uses nothing about v beyond that representation, so its bound (55) applies verbatim to $W^{(ij)}$ with the coefficients of (63): the two-point read of (38) differs from $W^{(ij)}$ by at most $\eta^{(ij)}$ at every $\tau \in [1, M-1]$, and every τ_ℓ lies there by hypothesis, so $|E_{ij}| = |\sum_\ell \varrho_\ell(\widehat{W}^{(ij)} - W^{(ij)})(\tau_\ell)| \le \eta^{(ij)} \sum_\ell \varrho_\ell$, the weights being positive. For the main term, $\tau_\ell = u_\ell/D = (\log n_\ell)/D$, hence $\omega_k\tau_\ell = \gamma_k \log n_\ell$ and $\cos(\omega_k\tau_\ell)+\mathrm{i}\sin(\omega_k\tau_\ell) = n_\ell^{\mathrm{i}\gamma_k}$ with $\gamma_k = \omega_k/D$; summing $\varrho_\ell W^{(ij)}(\tau_\ell)$ over ℓ and using $\mathrm{Re}[(a-\mathrm{i}c)(x+\mathrm{i}y)] = ax+cy$ gives the two groups of (65), the second carrying the extra factor $\tau_\ell = (\log n_\ell)/D$, which is what makes it the first logarithmic moment. That the two points are $2\pi/(ND)$ apart is $\omega_k = 2\pi k/N$.

For (66): the quoted inequality gives $|T_0(\gamma_k)| \le Q$ for each of the two k, and applied with the weights $\varrho_n \log n$, together with $\log n \le \log X \le 2\alpha = MD$, it gives $|T_1(\gamma_k)| \le MD\,Q$. Insert both into (65) and use the triangle inequality: the constant terms contribute at most $\sum_k(|a_k|+|c_k|)\,Q$ and the moment terms at most $\frac{1}{D}\sum_k(|b_k|+|d_k|)\,MD\,Q$, and the two sums are $\Lambda_0^{(ij)}Q$. □

On the reference window of Measurement 6.16 the two points and their spacing are $\gamma_1 = 0.6208$, $\gamma_2 = 1.2415$, $\Delta\gamma^{-1} = 1.611$, with $X = 2.465 \cdot 10^4$; they are read off (64) and not fitted.

Proposition 7.5 (five unconditional majorants)**.** *Let $\widehat{A}_2 = B - S$ be the split of Proposition 6.1, let $\mathcal{K}$ be the kernel of* (41)*, let ϱ be the weight vector and $\|\varrho\|_2$ its Euclidean norm. Assume $\widehat{a}_{11}\widehat{a}_{22} \ne 0$ and write $\Xi = |\widehat{a}_{11}\widehat{a}_{22}|$ for the length of this proposition only. Then, with $1 - r_{12}^2 = \det \widehat{A}_2/(\widehat{a}_{11}\widehat{a}_{22})$ by Lemma 5.8:*

- (R1) Cauchy–Schwarz on $\widehat{A}_2$. $|\det \widehat{A}_2| \le |\widehat{a}_{11}\widehat{a}_{22}| + \widehat{a}_{12}^2$, *hence* $|1 - r_{12}^2| \le 1 + \widehat{a}_{12}^2/\Xi \le 2$ *whenever $\widehat{A}_2 \succeq 0$. Hypotheses: none for the first inequality; $\widehat{A}_2 \succeq 0$ for the numerical bound* 2*.*

- (R2) Large sieve / duality on the comb–comb term. $|\det S| \le \|\mathcal{K}\|_{\mathrm{op}}\|\varrho\|_2^2$, *hence* $|1 - r_{12}^2| \le \big[|\det B| + |D(B,S)| + \|\mathcal{K}\|_{\mathrm{op}}\|\varrho\|_2^2\big]/\Xi$*. Hypotheses: $\mathcal{K}$ symmetric, which is Proposition 6.2; the operator-norm step is the Rayleigh bound and is the form in which the large sieve enters [18].*

- (R3) Montgomery–Vaughan on the three linear functionals. *On the prime-power instance with interior reads, Lemma 7.4 identifies the three reads as values of one exponential sum of length X, and of its first logarithmic moment, at the two points $\gamma_k = 2\pi k/(ND)$ of spacing $\Delta\gamma = 2\pi/(ND)$, and gives $|S_{ij}| \le \widetilde{Q}$ with*

$$\widetilde{Q} = \Lambda_0\,Q + \eta\|\varrho\|_1, \qquad Q^2 = (X + \Delta\gamma^{-1})\|\varrho\|_2^2, \tag{67}$$

  where $\Lambda_0 = \max_{ij}\Lambda_0^{(ij)}$ and $\eta = \max_{ij}\eta^{(ij)}$ are the coefficient budget and the interpolation bound of that lemma. Then $|1 - r_{12}^2| \le \big[|\det B| + (|B_{11}| + |B_{22}| + 2|B_{12}|)\,\widetilde{Q} + 2\widetilde{Q}^2\big]/\Xi$. Hypotheses: those of Lemma 7.4, the Montgomery–Vaughan inequality [18] at the two points that lemma derives, and the triangle inequality on $\det S = S_{11}S_{22} - S_{12}^2$*.*

- (R4) The exact split. $|1 - r_{12}^2| \le \big[|\det B| + |D(B,S)| + |\det S|\big]/\Xi$*. Hypotheses: the identity* (40) *and the triangle inequality; nothing else.*

- (R5) Vaughan Type I + Type II. *Write $\varrho = \sum_{q\le 4}\varrho^{(q)}$ for Vaughan's decomposition [24] of the weight vector and bound each of the sixteen resulting blocks of the bilinear form by its own operator norm as in* (R2)*. Then* $|1 - r_{12}^2| \le \big[|\det B| + |D(B,S)| + \sum_{q,q'}\|\mathcal{K}\|_{\mathrm{op}}\|\varrho^{(q)}\|_2\|\varrho^{(q')}\|_2\big]/\Xi$*. Hypotheses: those of* (R2) *for each block, plus Corollary 6.6*(iii)*.*

Routes (R2), (R3) *and* (R4) *are members of the class $\mathfrak{M}$ of Definition 6.5 as stated;* (R5) *is of the decomposed shape of Corollary 6.6*(iii) *with $p = 4$;* (R1) *is not a majorant of* $\det S$ *at all and is carried as the trivial baseline (Remark 6.7).*

Proof. (R1) is $|\widehat{a}_{11}\widehat{a}_{22} - \widehat{a}_{12}^2| \le |\widehat{a}_{11}\widehat{a}_{22}| + \widehat{a}_{12}^2$, and if $\widehat{A}_2 \succeq 0$ then $\widehat{a}_{12}^2 \le \widehat{a}_{11}\widehat{a}_{22}$. (R2): by (41), $\det S = \varrho^\mathsf{T}\mathcal{K}\varrho$, and for a symmetric matrix $|\varrho^\mathsf{T}\mathcal{K}\varrho| \le \|\mathcal{K}\|_{\mathrm{op}}\|\varrho\|_2^2$; insert into (40) and apply the triangle inequality. (R3): by Lemma 7.4 every $|S_{ij}| \le \widetilde{Q}$, so $|\det S| \le |S_{11}||S_{22}| + S_{12}^2 \le 2\widetilde{Q}^2$

and $|D(B,S)| \le (|B_{11}|+|B_{22}|+2|B_{12}|)\max_{ij}|S_{ij}| \le (|B_{11}|+|B_{22}|+2|B_{12}|)\widetilde{Q}$; insert into (40). (R4) is the triangle inequality on (40). (R5): expand $\det S = \sum_{q,q'}(\varrho^{(q)})^{\mathsf{T}}\mathcal{K}\varrho^{(q')}$ by bilinearity of D and bound each term as in (R2); that the four pieces sum to ϱ is Vaughan's identity, which is a coefficientwise statement about the weight vector and is verified as such (Section 8.3). Dividing by Ξ and using Lemma 5.8 gives each displayed form. □

Measurement 7.6 (the priced table)**.** Each majorant of Proposition 7.5 is evaluated window by window on the 67 node-complete windows of Surface II and its logarithm regressed against $\log h$. The resulting δ is the exponent that the route certifies for $1 - r_{12}^2 \le C\,h^{-\delta}$ on that surface: the bound is proved, the exponent is a fit, and the two are typed separately in every row.

route	where	δ	type of the row
$1 - r_{12}^2$ itself	Meas. 7.3	$+2.747$	measured (the object, not a route)
(R1) Cauchy–Schwarz on $\widehat{A}_2$	Prop. 7.5	-0.000	bound proved, exponent measured
(R2) large sieve / duality on $\det S$	Prop. 7.5	$+0.669$	bound proved, exponent measured
(R3) Montgomery–Vaughan on S_{11}, S_{22}, S_{12}	Prop. 7.5	-0.384	bound proved, exponent measured
(R4) exact split $\lvert\det B\rvert + \lvert D(B,S)\rvert + \lvert\det S\rvert$	Prop. 7.5	$+0.996$	bound proved, exponent measured
(R5) Vaughan Type I + Type II	Prop. 7.5	$+0.669$	bound proved, exponent measured

Row (R3) is priced on the bare Montgomery–Vaughan factor Q of (67), i.e. at $\Lambda_0 = 1$ and $\eta = 0$; the coefficient budget and the interpolation residual of Lemma 7.4 are window-dependent prefactors of that factor, and the exponent in the row is the exponent of Q on this surface. Every other row is priced on the majorant exactly as displayed.

The best of the five on this surface is the triangle inequality on the exact three-term split, at $\delta = +0.996$; the exponent in (60) is 3, so on this surface the best priced route falls short by 2.004 *in the exponent.* Two readings are part of the measurement. Corollary 6.6(ii) says why the three norm-based rows cannot do better in principle: they see the image point $y(S) = (S_{11}, S_{22}, S_{12})$ and not its relation to B, and the phenomenon of Measurement 6.16 is exactly that relation. Theorem 6.3(v) says why (R2) and (R5) land on the same number: a bilinear tool applied to a kernel of rank three degenerates to a statement about three linear functionals, and the Vaughan blocks inherit the degeneracy block by block.

Measurement 7.7 (the structural obstructions)**.** Five further routes are excluded not by their exponent but by their shape. Each is exhibited failing on the windows of Surface I (Section 8.2), 12 windows with $m \le 300$, in the way the classification predicts.

(N1) *The symbol infimum is negative.* The Toeplitz symbol $\sigma(\theta) = c_0 + 2\sum_{d\ge1} c_d \cos(d\theta)$ has $\min_\theta \sigma \in [-47.6, -21.3]$ on 12/12 windows, while the parity compression is positive definite there. No pointwise minorant of the symbol can therefore deliver a positive floor: the positivity is a finite-section fact — a Fejér cancellation [7] between the Toeplitz and the Hankel half of (2) — and not a symbol fact.

(N2) *Perturbation theory prices at the wrong scale.* Kato's bound $|\nu_2(\widehat{A}_2 + E) - \nu_2(\widehat{A}_2)| \le \|E\|$ [16] is sharp at the scale of the matrix, and on Surface I the small eigenvalue sits a factor $9.8 \cdot 10^2$ to $2.0 \cdot 10^4$ below $\|\widehat{A}_2\|$. A perturbative argument would have to control the matrix to that relative accuracy first, which is the original problem. The observation is sharper than a divergence statement: on a declared family of perturbative expansions the Neumann/Jacobi series diverges (radius 1.0 to 12.7) while the Kato series *converges*, with radius 0.067 — to the wrong object, since the exact bottom eigenvector of the corresponding 6×6 block overshoots the constrained minimum by a relative excess 6.04 at overlap 0.083 and relative gap $5.4 \cdot 10^{-7}$.

(N3) *Band-local domination fails.* On the mode-1 weight vector the comb half of the lag sum dominates the archimedean half by a factor 1.71 to 2.83 on 12/12 windows, so a route that would dominate the comb band by the archimedean band loses on the instance.

(N4) *The signed sum is far below its ℓ^1 mass.* On the declared subsample of the reference window the ratio is 40.6 (Measurement 6.8, equation (47)); any route that replaces the signed double sum by its absolute value pays that factor before it starts.

(N5) *The L_P control is exact and empty.* Replacing the comb by nothing — i.e. running the whole reduction on the lag sequence c^L of Lemma 2.2 — gives $\widehat{a}_{12} = 1.1 \cdot 10^{-17}$, hence $1 - r_{12}^2 = 1$ with no decay at all at $m = 64$, 128 and 256, a Gram matrix that is diagonal to an off-diagonal mass $2.3 \cdot 10^{-13}$, and a ladder gain $g_K/g_1 = 1$ at zero tolerance. Any argument that would already work for L_P alone is therefore vacuous for A: the phenomenon of Problem 7.1 is not a property of the two parity modes.

7.4 The density curve

The measurements so far are taken along a ladder in h at a fixed rule relating the resolution to the window. The next one is taken on a rectangle, which separates the window index from the number of comb nodes per lag cell, and it is the last decidable measurement the declared caps permit.

Definition 7.8 (the ratio and its binned exponent)**.** For a window put

$$\gamma = \frac{\lambda_{\min}(G)}{\lambda_{\max}(G)}, \qquad R = \frac{\gamma}{1 - r_{12}^2}, \qquad \mathrm{dens} = \frac{\#\{\ell : u_\ell \le 2\alpha\}}{M}, \tag{68}$$

with G the normalised K-block of (7) at $K = 16$ and dens the number of comb nodes per lag cell. For a density interval $[\lambda, 4\lambda)$ the *binned exponent* is: for each anchor with at least three ladder rungs inside the interval, the ordinary least-squares slope of $\log |R|$ against $\log h$ over its rungs; then the unweighted mean of the anchor slopes, with the cluster-robust standard error of that mean, and with the sign reversed so that a positive value means that R falls. The procedure is deterministic and independent of the order in which the cells are enumerated.

Two remarks fix the reading. First, R compares two spectral functionals of one and the same finite matrix — the relative spectral gap of the normalised low block, and the near-degeneracy scalar of Problem 7.1 — so a positive binned exponent says that the second closes more slowly than the first, on that surface, at that density. Second, the binned exponent is a *local slope of a curved surface* and not an exponent of a power law: $\log R$ is measurably curved in $\log h$ (the per-anchor quadratic coefficient is -0.0842 ± 0.0085), which is why the value depends on the rung ladder — see Remark 7.10.

Measurement 7.9 (the density curve)**.** Surface III of Section 8.2: 692 anchors, 14 532 windows, a node table capped at $2.5 \cdot 10^7$ containing 1 566 727 nodes, every window with a complete comb, none of them indefinite. The cap and the smallest admissible rung force dens ≤ 6120, of which 6118 is attained; the attained density spans 4.0 decades. On the declared geometric ladder of ratio-4 density bins, fixed before any window was built, the binned exponent of Definition 7.8 is:

bin (nodes per lag cell)	cells	anchors	exponent $\pm$ s.e.	on the $\sqrt{2}$ sub-ladder
[0.1, 0.4)	83	11	$+0.8886 \pm 0.1027$	$+0.9818 \pm 0.2129$
[0.4, 1.6)	318	34	$+0.5148 \pm 0.0883$	$+0.5762 \pm 0.0743$
[1.6, 6.4)	597	69	$+0.1948 \pm 0.0861$	$+0.2073 \pm 0.0679$
[6.4, 25.6)	1056	117	$+0.3922 \pm 0.0935$	$+0.1942 \pm 0.0734$
[25.6, 102.4)	1924	218	$+0.1520 \pm 0.0573$	$+0.0879 \pm 0.0589$
[102.4, 409.6)	3588	403	$+0.0579 \pm 0.0542$	-0.0573 ± 0.1320
[409.6, 1638.4)	4891	471	$+0.0376 \pm 0.0410$	—
[1638.4, 6553.6)	2075	272	$+0.1046 \pm 0.0881$	—
pooled, the two densest bins			$+0.0496 \pm 0.0372$	—

Every band is the cluster-robust standard error of Definition 7.8, the cluster being the anchor; the pooled row is the inverse-variance pool of the last two bins. Read against zero, the eight bins sit at 8.7, 5.8, 2.3, 4.2, 2.7, 1.1, 0.9 and 1.2 standard errors, and the pool of the last two at 1.3. The joint slope of the binned exponent against log(dens) is -0.0725 ± 0.0090, i.e. 8.0σ over the 4.0 attained decades. The violation of monotonicity in the fourth bin is declared and not smoothed; the last column is where it comes from.

That column is the same bins evaluated on the $\sqrt{2}$ sub-ladder of the same rung ladder, over the anchor range of the earlier and 20.8 times smaller node table, and it carries two readings at once. It is the *node-table* consistency check: each of the six entries reproduces the value published for that bin under the smaller table to 0.0σ, which is the sharpest form the check can take — with the caveat that the sixth bin ran to 500 rather than to 409.6 there, so the comparison is like-for-like only up to that edge. And it is the *estimator* dependence, exhibited rather than asserted: the two exponent columns differ by up to 0.20, the largest gap in the fourth bin and the next in the sixth. What differs between them is the rung ladder and the anchor range; what does not is the node table, so the movement is the estimator and its sample and not the arithmetic input. That is the caveat of Remark 7.10 as a number.

Two further consistency facts belong to the measurement. *(i)* A larger node table adds nodes above the old cap and nothing else, so on every window whose comb was already complete under the old cap the lag vector is unchanged bit for bit, R is unchanged exactly, and the binned exponent is identical — a statement about complete combs, not about tables, and its must-break control is a window whose comb was truncated under the old cap, which does move. *(ii)* The two densest bins have no counterpart under the smaller table at all: they lie beyond the density its cap could reach, which is why they are the reason for the enlargement.

Remark 7.10 (what the curve does and does not decide, and at what price)**.** Three statements are part of the claim rather than caveats attached to it.

Undecidability. On this surface, a true zero of the binned exponent at finite density, a power-law approach to zero, and a plateau at any level below 0.074 are mutually indistinguishable: 0.074 is the two-sigma resolution of the densest bins, and no finite node table reachable under the declared budget separates the three. No limit is claimed, and the quantifier in Problem 7.1 is untouched by the curve.

The resource fence. The resolution narrows like the square root of the anchor cap, so the width of the plateau window improves by a factor 2 for every factor 4 in the node table; raising the table by the factor 20.8 that produced Surface III narrowed the window from 0.264 to 0.074, a factor 3.6. Extending the curve by one further decade of density needs a node table of about $2.5 \cdot 10^9$, a factor 100 above the one used; that is outside the declared budget, and — this is the honest half — it is not obviously the right question either, since the object being extrapolated is a local slope of a curved surface and not an exponent.

The ladder caveat. The estimator of Definition 7.8 is ladder-dependent. The same density bin evaluated on a $2^{1/4}$ rung ladder and on its $\sqrt{2}$ sub-ladder gives different values, with shifts up to 0.20 on the surface above, because a bin with about eight rungs samples the curvature of $\log R$ differently from one with about four. Every number in Measurement 7.9 is therefore quoted *with* its rung ladder, and a bin value quoted without one is not reproducible.

7.5 Placement: two controlled experiments and one identity

The results of Sections 2–6 hold for an arbitrary finite comb (Remark 2.1). It is therefore a fair question where the phenomenon of Problem 7.1 actually lives. Two interventions answer it, and the second one closes into an identity.

Measurement 7.11 (the value-multiset scramble)**.** Replace the node positions u_ℓ by sorted uniform samples on the same range, keeping the weight multiset $\{\varrho_\ell\}$ exactly, and recompute. On 60 declared draws over the windows of Surface I:

- the rank-three collapse is untouched: $\sigma_4/\sigma_1 \le 2.5\cdot 10^{-16}$ on every draw, as Theorem 6.3(iii) requires — the theorem does not know where the nodes are;
- every majorant of Proposition 7.5 moves by a bounded factor and keeps its exponent;
- the *value* $1 - r_{12}^2$ moves by a median factor 679, range 26 to $4.9 \cdot 10^3$. An independent set of draws in the rank module gives median 835, range 69 to $1.8 \cdot 10^4$; both are quoted, and the spread between them is the spread of the sampling.

On a second declared surface the same intervention destroys positive definiteness of the low block on all 24 draws and moves R by a median factor $\approx 10^6$; on the first of the two new density bins of Surface III it destroys it on 194/194 cells with the cell occupancy preserved exactly, so that only the sub-cell placement is randomised. On Surface II the decay is removed outright: over 18 windows the fitted exponent of $1 - r_{12}^2$ moves from -2.747 to $+0.009$, a gap of 2.76 in the exponent; on a further family the angle $\sin\angle$ rises by a median factor $6.6 \cdot 10^4$ and its exponent moves from -1.93 to $+0.11$. Position and weight are both load-bearing, and the second is checked separately: keeping the node positions and replacing each weight by the mean weight gives $h^{-0.116}$ against the true $h^{-2.747}$. The conclusion is the one sentence this subsection exists for: *every theorem in this paper survives the scramble and the phenomenon does not*, so the theorems are provably not where the arithmetic sits. What moves is the joint value of (S_{11}, S_{22}, S_{12}) against B — exactly the object Corollary 6.6(ii) says no member of $\mathfrak{M}$ can see.

The second experiment displaces a few nodes by a fraction of a lag cell. That the response is available in closed form is not an accident: the assembly is affine in the fractional positions.

Lemma 7.12 (the exact placement derivative)**.** *Fix a finite set F of nodes and displace their fractional positions, $\theta_\ell \mapsto \theta_\ell + \vartheta$ for $\ell \in F$, keeping m_ℓ and all other nodes fixed. As long as no $\theta_\ell + \vartheta$ leaves $[0,1)$, the lag sequence is affine in ϑ,*

$$c(\vartheta) = c(0) + \vartheta\, \dot c, \qquad \dot c = \tfrac{1}{2} \sum_{\ell\in F} \varrho_\ell (\delta_{m_\ell} - \delta_{m_\ell+1}), \tag{69}$$

where δ_m is the lag sequence supported at the single lag m. In particular $\dot c$ is exact and piecewise constant in ϑ, and the induced $\dot A = \Pi^{\mathsf T} T_M(\dot c)\Pi$, $\dot G = \mathcal{M}^{-1/2}(t_i^{\mathsf T} \dot A t_j)_{ij}\mathcal{M}^{-1/2}$ are exact as well.

Proof. By the two-point read in the proof of Proposition 6.1, the node u_ℓ contributes $-\varrho_\ell \frac{1}{2}[(1-\theta_\ell)\delta_{m_\ell} + \theta_\ell \delta_{m_\ell+1}]$ to c and nothing else. This is affine in θ_ℓ with slope $\frac{1}{2}\varrho_\ell(\delta_{m_\ell} - \delta_{m_\ell+1})$; summing over $\ell \in F$ gives (69). The reflected hat still vanishes and m_ℓ is unchanged because $\theta_\ell + \vartheta$ stays in $[0,1)$, so no other lag entry moves. The maps $c \mapsto T_M(c) \mapsto \Pi^{\mathsf T} \cdot \Pi$ and $A \mapsto G$ are linear, which transports exactness to $\dot A$ and $\dot G$. $\square$

Proposition 7.13 (the placement response in closed form)**.** *Let $\vartheta \mapsto G(\vartheta)$ be as in Lemma 7.12, suppose $\lambda_{\min}(G)$ and $\lambda_{\max}(G)$ are simple with unit eigenvectors v and w, and suppose $\widehat a_{11}\widehat a_{22} \neq 0$*

and $r_{12}^2 \neq 1$. Then, with $R = \gamma/(1 - r_{12}^2)$ as in (68)*,*

$$\frac{\mathrm{d}\log R}{\mathrm{d}\vartheta} = \frac{v^\top \dot{G} v}{\lambda_{\min}(G)} - \frac{w^\top \dot{G} w}{\lambda_{\max}(G)} - \frac{1}{1 - r_{12}^2}\,\frac{\mathrm{d}\,(1 - r_{12}^2)}{\mathrm{d}\vartheta}, \tag{70}$$

where the last derivative is the closed expression

$$\frac{\mathrm{d}\,(1 - r_{12}^2)}{\mathrm{d}\vartheta} = -\frac{2\widehat{a}_{12}\dot{a}_{12}}{\widehat{a}_{11}\widehat{a}_{22}} + \frac{\widehat{a}_{12}^2}{\widehat{a}_{11}\widehat{a}_{22}}\Big(\frac{\dot{a}_{11}}{\widehat{a}_{11}} + \frac{\dot{a}_{22}}{\widehat{a}_{22}}\Big), \qquad \dot{a}_{ij} = t_i^\top \dot{A} t_j. \tag{71}$$

Every quantity on the right is an explicit finite sum; no fit enters.

Proof. For a simple eigenvalue of a symmetric matrix depending differentiably on a parameter, $\mathrm{d}\lambda/\mathrm{d}\vartheta = v^\top \dot{G} v$ with v the normalised eigenvector [13, 8]; by Lemma 7.12 the dependence is affine, so $\dot{G}$ is constant and the hypothesis of differentiability is automatic. Then $\mathrm{d}\log\gamma/\mathrm{d}\vartheta = \dot{\lambda}_{\min}/\lambda_{\min} - \dot{\lambda}_{\max}/\lambda_{\max}$, and $\log R = \log\gamma - \log(1 - r_{12}^2)$ gives (70). For (71), differentiate $1 - r_{12}^2 = 1 - \widehat{a}_{12}^2/(\widehat{a}_{11}\widehat{a}_{22})$ by the quotient rule; the entries $\widehat{a}_{ij}$ are affine in ϑ with derivatives $\dot{a}_{ij}$. □

Measurement 7.14 (causality, the pole, and where the response lives)**.** Three numbers on a declared instance family.

(i) The intervention is causal, and linear over four decades. Displacing six declared nodes inside their lag cells — the cell index fixed, the mass exactly preserved, twelve lag entries changing — gives $\mathrm{d}\log R/\mathrm{d}\vartheta = 879$, linear to a factor 1.011 over the declared range $\vartheta = 10^{-7}$ to 10^{-4}, with an exact $\vartheta = 0$ rebuild control (residual 0 on 40 windows). This is a statement of causation, not of correlation: nothing but the placement was changed.

(ii) The number is an identity, not a regression. On the declared instance list the closed form (70) reproduces the full nonlinear rebuild to a median relative residual $3.37 \cdot 10^{-6}$ at the minimum of a declared central-difference step ladder, with the U-shape exhibited — round-off limited below, truncation limited above, both ends at least three times the minimum. A wrong first-order formula has a floor, not a minimum; two deliberately mutated formulas (swapped eigenvectors, dropped last term) miss the measured quotient by a factor of order 40. Of the three terms of (70) the last one, the near-degeneracy channel, carries the largest share on every instance — a median 84%, against 7% for the $\lambda_{\min}$ term.

(iii) The intervention walks along a near-degeneracy that is already there. Sweeping the full placement period on a declared 64-point grid, positive definiteness of the low block survives only about 2% of the period: $\lambda_{\min}$ changes sign inside the period, so $\log R$ has a *pole* there and is not defined on the whole period. The bisected loss point satisfies $\vartheta_{\mathrm{PD}} \cdot |\mathrm{d}\log R/\mathrm{d}\vartheta| = 5.8$, a pole at the natural scale, with $\vartheta_{\mathrm{PD}} = 5.9 \cdot 10^{-3}$ median. And the two lowest modes of the normalised low block already sit only 13.4% apart at $\vartheta = 0$. Consequently a first-harmonic regression of $\log R$ over the period estimates a Fourier coefficient of a function with a pole in the interval, and returns 2.89 against a local derivative of 879; neither number is wrong, they measure different things, and the discrepancy is not an anomaly but a pole.

The cell phases themselves are equidistributed in the classical sense: they are generated by $\varphi_\ell(h) = \{h\,\theta_\ell\}$ from one real invariant per node — an identity certified to $1.14 \cdot 10^{-13}$ over 1316 windows and eight nodes — and the largest circular-uniformity statistic over those windows is 1.41, i.e. consistent with uniformity after correction for multiplicity. Equidistribution of $\{h\theta\}$ is Weyl's theorem [26]; it is cited as the classical address of the phases and is used in no proof.

7.6 What is not claimed

Remark 7.15 (what is not claimed)**.** Problem 7.1 is open. Nothing in this paper bounds it, and nothing in this paper is conditional on it. Explicitly:

(i) No statement of this paper is an arithmetic statement. Theorems 3.1, 4.1, 6.3 and every proposition, lemma and corollary of Sections 2–6 hold for an arbitrary finite comb, and Measurement 7.11 is the controlled experiment that says what changes when the comb does.

(ii) No fitted exponent is used as a bound. In particular the δ column of Measurement 7.6 certifies nothing beyond the finite surface it was fitted on; the inequalities of Proposition 7.5 are the proved content of those rows.

(iii) Measurement 7.9 claims no limit. Its explicit content includes its own undecidability (Remark 7.10), and the quantifier in (60) is untouched by it.

(iv) Positivity of A on the parity sector is used only where it is named as a hypothesis, and there only as a per-window certificate. Lemma 5.8 exists so that the reduction never needs it.

(v) No comparison of a demanded precision against the strength of any conjecture appears anywhere in this paper. Problem 7.1 is stated as a question about a finite bilinear form and about nothing else.

8 Numerical verification

Every proved statement of this paper is recomputed by a machine check, and every measured number is quoted with the surface it was taken on. This section states the convention under which a numerical statement counts as a certificate, declares the three surfaces on which anything is evaluated, and gives the table that maps each statement to its check.

8.1 Floating-point convention and declared floors

Positivity statements are completed Cholesky factorisations relative to a declared floor in the sense of Wilkinson and Higham [27, 14]: with $u = 2^{-53}$ the unit roundoff and $c_h = (h+1)/(1-(h+1)u)$, a completed factorisation of $X - \delta I$ with $\delta = c_h u\|X\|_\infty$ certifies $\lambda_{\min}(X) > 0$, and the cheap certified bound $\|X\|_2 \leq \|X\|_\infty$ is used for the norm [9]. Quantities obtained from an eigenvalue routine are labelled *measured* and never used as certificates. Diagonally dominant comparisons, where used, are Ostrowski's [19].

Three further conventions are fixed once.

(F1) *Near-null horizons are declared per window, not globally.* An identity between two quantities that cancel to k digits is checked against a bar proportional to u times the larger of the two halves, computed on that window. Where the paper quotes a residual — for instance $1.0 \cdot 10^{-10}$ for the polarisation (40) — the bar it is compared against is quoted with it. The cancellation reaches seven digits on the deepest windows ($|1 - r_{12}^2|$ down to $9.3 \cdot 10^{-8}$), and that residual sits an order of magnitude below the declared near-null horizon 10^{-9} for that identity. Identities that do not cancel are checked at the ordinary round-off scale and their residuals are reported there.

(F2) *Conditioning cap.* No matrix with cond $\geq 10^{12}$ is factorised or inverted anywhere; windows exceeding the cap are dropped before any number is read from them, and the number dropped is reported with the surface.

(F3) *Every identity carries a mutation control.* A deliberately wrong variant of the same identity — a swapped eigenvalue pair, a dropped term, an off-by-one in the lag index — is evaluated alongside it and must fail by a margin that is also reported. An identity verified without a mutation control is not counted.

8.2 The three surfaces

Three surfaces are declared and never mixed; each is fixed before any number is read from it.

Surface I (exact, small). The 12 deepest prime-power anchors admitting a window inside the lag cap, $n = 29$ to 139, $m = 96$ to $291 \leq 300$. Every identity of Sections 2–6 is recomputed from scratch here, together with the structural obstructions of Measurement 7.7. The one unconditional arithmetic input, $|\psi(x) - x| \leq \kappa x$ with $\kappa = 0.038821$, is verified at every jump point of $\psi(x)/x$ on the finite table [4, 20].

Surface II (measurement). The frame-A family with $h = 142$ to 1445, all windows with $\mathrm{cond}(G) \leq 10^{12}$. Two subsets are used and are always named. The *ladder*, 27 windows with $h = 142$ to 1320, carries Measurements 3.6, 4.5 and 6.16. The *refinement*, 70 windows with $h = 142$ to 1445, carries Measurements 7.3 and 7.6; of these, 67 have a comb complete under the node cap and span $h = 142$ to 1433, and every rate quoted on the node-complete subset is quoted alongside the value on the full family (Measurement 7.3). Truncation of the comb by the node cap is not a nuisance to be averaged away: on a truncated window the low block can be indefinite, with $\lambda_{\min} = -9.6 \cdot 10^4$ on the declared stress instance, while every window with a complete comb on this surface is positive definite. Indefiniteness sits at the declared sieve horizon and nowhere else.

Surface III (the density rectangle). A rectangle, not a ladder: every prime power in the declared anchor range is an anchor — 692 of them — crossed with a rung ladder of ratio $2^{1/4}$ containing the seven rungs of the earlier ladder verbatim, 21 rungs from $h = 128$ to $h = 1280$. This gives 14 532 windows, all with a complete comb and none indefinite. The node table is capped at $2.5 \cdot 10^7$ and contains 1 566 727 prime powers. The cap and the smallest admissible rung together force dens ≤ 6120, of which 6118 is attained; the smallest rung is itself forced, since a $K \times K$ low block needs $2K$ modes and the conditioning floor of (F2) does the rest. Surface III carries Measurement 7.9 and the sweeps of Measurement 7.14. The bin edges, the rung ladder and the anchor range are all fixed before the first window is built; this is the anti-fitting device of the surface, and Remark 7.10 is the price it charges.

8.3 Cross-reference table

Three readings are quoted in this paper and are not in the table below, because no module recomputes them: the tightness range of Measurement 3.6, the fitted exponents of Measurement 7.3 and the four numbers of Measurement 6.16. Each is a quantity read off an eigenvalue routine on a surface declared above, each is typed *measured* in Table 1, and none is used as a bound anywhere; what the modules certify are the identities those readings are readings of. The definitions and remarks are likewise absent: a definition is a procedure and a remark is a reading, and both are typed as such in Table 1.

Table 3: Every proved statement and every measurement carried by a module, with the check that recomputes it. Module identifiers refer to the verification suite accompanying this work; item numbers are those of the module documentation.

statement	check	what the check recomputes
Lem. 2.2 (parity spectrum)	`v555`–`v559`, parity control	$L_P t_k = \mu_k t_k$ with an orthonormal basis; $\Pi^{\mathsf{T}} T_M(c^L)\Pi = L_P$ at zero tolerance

continued on the next page

statement	check	what the check recomputes
Lem. 2.3, 2.4 (interlacing, nesting)	`v559` links 1, 4	the two-block Schur floor; a K-truncated Rayleigh minimum as an upper bound for $\lambda_{\min}$ and never a lower one
Lem. 2.9 (gauge law)	`v555` item (4); `v559` link 3	$\sum_d w_d = 0$; $\sum_d c_d w_d = x^{\mathsf{T}} G x$
Lem. 2.10 (Abel swap)	`v555` item (4); `v559` link 9	the boundary-free identity at levels $k = 1, 2, 3$
Prop. 2.6 (three ladder forms)	`v557` item (1) I1–I3; `v559` link 12	prefix from one Cholesky; minor form; regression form
Lem. 2.7 (diagonal invariance)	`v557` item (2)	invariance of g_K/g_1 under $G \mapsto \Lambda G \Lambda$
Prop. 2.8 (ladder direction)	`v556` item (5); `v559` link 4	$s \geq g_K \geq g_1$ with strictly positive increments
Thm. 3.1 (TV floor)	`v555` item (2), steps T1–T3	$w_0 = \|a\|^2$; $\mathrm{TV} \geq \|w_0\|$; $\mu_1 \,\mathrm{TV} \geq 1$
Cor. 3.3 (Abel/Hölder)	`v555` item (4); `v559` link 9	$\|Q\| \leq \mathrm{TV} \max_d \|C_d\|$
Cor. 3.4 (floor at one coordinate)	`v556` item (4b)	$\mathrm{TV} \geq \|v\|^2 \geq (t_1^{\mathsf{T}} v)^2$
Cor. 3.2 (the price)	`v555` item (2), step T3	(19) is $\mu_1 \,\mathrm{TV} \geq 1$ divided by TV, and that inequality is what the step certifies
Rem. 3.5 (negative control)	`v555` item (2), control	an inadmissible vector undercuts the floor
Thm. 4.1 (gauge)	`v556` item (1); `v559` link 10	degree-two homogeneity; ratio equal across sectors
Prop. 4.3, Cor. 4.2 (conservation, no escape)	`v556` item (2)	floor $\times$ transfer $= 1/\mu_1$ at every shift of the declared shift list, which is also the corollary
Meas. 4.5 (the sector battery)	`v556` item (1)	the ratio across the declared battery of weight families, taper widths and shifts; the wrong transport map failing loudly

continued on the next page

statement	check	what the check recomputes
Prop. 4.4 (null mode, other sector)	`v555`–`v559`, parity control	the sine family at $k \geq 1$ with an orthonormal basis; its parts (ii) and (iii) are closed forms of the same recurrence
Prop. 5.1 ($K = 2$ exact)	`v557` item (3); `v559` link 13	the closed vector attains $1/g_2$; $G_{11}g_2 = 1/(1-r_{12}^2)$
Prop. 5.2, 5.4 (pivot, entrywise)	`v557` item (4)	the pivot identity; $\lvert x^{\star\mathsf{T}} E x^\star \rvert = \varepsilon\Theta_K$
Lem. 2.5 (trial direction)	`v557` item (5); `v559` links 4, 7	every admissible trial is a lower bound for $1/g_K$
Lem. 5.6, 5.7 (Wronskian, minors)	`v558` item (1) I3	the telescope with $W_{-1} = W_{h-1} = 0$; $\frac{1}{2}\sum \mathcal{W}^2 = 1/(\mu_1\mu_2)$
Lem. 5.8 (determinant reading)	`v558` items (2), (6); `v559` link 14	the three equal expressions, no positivity used
Lem. 5.9 (Gershgorin)	`v558` item (2)	$\nu_1 \leq \max(\widehat{a}_{11}, \widehat{a}_{22}) + \lvert\widehat{a}_{12}\rvert$
Prop. 5.10 ($t^\star$ collapse)	`v558` item (3)	the closed form (36); the harmonic-mean bound
Prop. 6.1, 6.2 (split, polarisation)	`v558` item (4) TH1–TH3; `v559` link 15	$\widehat{A}_2 = B - S$; (40); the kernel $\mathcal{K}$
Thm. 6.3 (rank three)	`v558` item (5) TH4; `v559` link 15	J, its spectrum, $\operatorname{rank}\mathcal{K} \leq 3$, the inertia bound
Lem. 6.4, Meas. 6.8 (wedge, ℓ^1)	`v558` items (4), (6)	the wedge reading, and its rank-one defect against the declared bar $5 \cdot 10^{-2}$; the ℓ^1-to-signed ratio itself is measured on the declared subsample and is not recomputed
Cor. 6.6 (majorants)	`v558` item (6)	the image of a norm ball; the scramble discriminator

continued on the next page

statement	check	what the check recomputes
Lem. 6.9–6.12 (weights, band, spline)	`v558` item (5), band checks	(48) at zero tolerance on $1 \le d \le M-1$; the fit of (w_d) in (52); the interpolation bound (56)
Prop. 6.13, Meas. 6.15 (Type II)	`v558` item (5) TH5	$\sigma_{4K+1} \le \eta\sqrt{RC}$ on an interior block; the measured singular-value profile against a generic kernel
Prob. 7.1 (the open problem)	`v559` link 16	the *shape* is certified per window; the rate and the quantifier stay open
Lem. 7.2 (angle against determinant)	`v558` item (2)	the ratio (61) and its two-sided bound (62)
Lem. 7.4 (dual-point read)	`v558` items (4), (5)	the three reads as two-point reads of the polarised closed weights against the direct block; those weights seeing a node only through $\log n$; the parity frequencies ω_k at zero tolerance, the two points of (64) being ω_k/D
Prop. 7.5 (five majorants)	`v558` items (5), (6); `v559` NG1–NG8	each majorant re-derived from its hypotheses; Vaughan's identity coefficient by coefficient
Meas. 7.6 (priced routes)	`v559` NG1–NG8	each classified route exhibited failing as classified
Meas. 7.7 (structural obstructions)	`v559` NG1–NG8; `v558` item (6)	the symbol infimum; the Kato scale and the two radii; the band-local ratio; the L_P control at zero tolerance

continued on the next page

statement	check	what the check recomputes
Meas. 7.9 (the density curve)	`v562` items (3), (4)	the binned exponent as a deterministic procedure; the ladder dependence; the node-table invariance on complete combs, with its must-break
Lem. 7.12, Prop. 7.13 (placement response)	`v562` item (1)	$\dot{c}$ exact and piecewise constant; (70) against a full rebuild on a step ladder with its U-shape and two mutants
Meas. 7.14 (causality, the pole)	`v562` items (1), (2)	the slope and its linearity; the sign change inside the period; the near-degeneracy at $\vartheta = 0$
Meas. 7.11 (the scramble)	`v558` item (6); `v560` item (5); `v562` item (2)	rank and majorants unchanged; the value moved; the low block lost at fixed cell occupancy

8.4 Two remarks on the fits

Remark 8.1 (the floor exponent against the closed exponent)**.** Measurement 3.6 reports the floor of Theorem 3.1 growing as $h^{+1.997}$ on Surface II, while (6) gives the closed value $1/\mu_1 = 1/(4\sin^2(\pi/N))$ with $N = 2h+1$, whose growth is exactly h^2 up to the discrete correction. The two agree, and the small deficit is that correction: expanding the sine, $1/\mu_1 = (2h+1)^2/(4\pi^2)\,(1 + O(h^{-2}))$, so a least-squares fit of $\log(1/\mu_1)$ against $\log h$ over a finite window range returns $2 - c/\log h_{\text{eff}}$ for a positive c coming from the $+1$ in $2h+1$. The fit is therefore a consistency check on the surface and not an independent statement; the closed form is what is used, and the fitted number is quoted because suppressing it would hide the size of the discretisation.

Remark 8.2 (two reported exponents, one measured object)**.** Measurement 7.3 reports $h^{-2.747}$ for $1 - r_{12}^2$ and $h^{-2.711}$ for the sine of the angle between the rows of $\widehat{A}_2$, on the same 67 windows. These are not two independent readings. By Lemma 7.2 the two quantities differ by the factor $\widehat{a}_{11}\widehat{a}_{22}/(\rho_1\rho_2)$, which lies in $(0, 1]$ and equals $\frac{1}{2}$ times the ratio of the geometric to the arithmetic mean of the two diagonal entries when r_{12}^2 is close to 1; the diagonal ratio $\widehat{a}_{22}/\widehat{a}_{11}$ stays in a bounded interval on the surface, so the factor is bounded above and below there and can shift the fitted exponent only by the slow drift of that ratio. The observed difference, 0.036 in the exponent, is of exactly that size. The two numbers are reported because both readings of Problem 7.1 are in circulation and a reader should be able to match either; no regression theorem is claimed, and neither number is used as a bound.

8.5 Reproducibility

The checks are grouped into eight modules, `v555` to `v562`, run in a single suite. Each module rebuilds its surface from the declared caps rather than reading stored data, verifies the arithmetic input at every jump point of the finite table it uses, and carries the mutation controls of (F3). Total runtime is under two seconds for the identity modules and about two minutes for the module that builds Surface III. Every matrix that is factorised or inverted anywhere in the suite satisfies the cap of (F2); the two exceptions — the truncated-comb stress instance of Surface II and the deliberately inadmissible vector of Remark 3.5 — are must-break controls, are declared as such, and no number is read from them except the failure they are built to exhibit.

9 Limitations, and what is not claimed

(L1) **No uniformity in the window index.** Every statement typed *certified per window* in Table 1 is verified window by window on a declared finite surface, and no quantifier over windows is claimed. In particular the hypothesis of Proposition 2.8 and the positivity hypothesis of (37) are certificates, not theorems.

(L2) **Positivity facts are facts about finitely many finite matrices.** That A is positive definite on the parity sector on the tested windows is a numerical statement with a declared floor. It is not routed through, compared with, or used to suggest anything about any criterion; Lemma 5.8 exists precisely so that the reduction never needs it.

(L3) **Fits are fits.** Every exponent in Measurements 3.6, 7.3, 7.6 and 7.9 is a least-squares fit on a declared surface, reported for orientation. No fit is used as a bound anywhere. In particular the δ column of Measurement 7.6 is a property of the surface: what is proved about those five routes is the inequalities of Proposition 7.5, and nothing else.

(L4) **The exponent is not one number.** On the surfaces of Measurement 7.3 the fitted exponent of $1 - r_{12}^2$ ranges from -1.27 to -2.95, and the two readings of Problem 7.1 give different fits on the same windows (Remark 8.2). Any sentence of the form "the quantity decays like $h^{-\delta}$" is a statement about a declared surface and a declared reading, and this paper never makes it in any other form.

(L5) **The dense corner is exhausted only up to the declared caps.** Measurement 7.9 states its own undecidability, and that statement is part of the claim: a true zero, a power-law approach, and a plateau below 0.074 are indistinguishable on Surface III. Extending the curve by one decade costs a node table a factor 100 larger, which is outside the declared budget.

(L6) **The curve estimator is ladder-dependent.** The binned exponent of Definition 7.8 shifts by up to 0.20 between the declared rung ladder and its coarse sub-ladder, because $\log R$ is curved in $\log h$ (measured per-anchor curvature -0.0842 ± 0.0085). Every value in Measurement 7.9 is quoted with its rung ladder, and a value quoted without one is not reproducible. A ladder-robust estimator — integrating the curvature rather than sampling it — is not attempted here.

(L7) **Not every standard error is reproduced.** For the six older bins of Measurement 7.9 the source reports the values and the joint density slope but not the individual standard errors, and the table says so rather than filling the column. The declared monotonicity violation in the fourth bin is likewise reported and not smoothed.

(L8) **One phrase did not convert.** The bandwidth $\Rightarrow$ η-rank mechanism is proved (Lemmata 6.9–6.12, Proposition 6.13), but it delivers $4K$ singular values at a declared truncation error and only for blocks satisfying the interior hypothesis (57); it does not deliver "effective rank $O(1)$", and the smaller number actually observed stays a measurement (Measurement 6.15). Remark 6.14 states what the interior hypothesis excludes.

(L9) **The comb's arithmetic is not used by any theorem.** Remark 2.1 is meant literally: replacing the prime-power instance by an arbitrary finite comb changes no proof in Sections 2–6. What changes is the *value* of the three functionals of Theorem 6.3(v), and Measurement 7.11 is the controlled experiment that says so.

(L10) **The placement results are local.** Proposition 7.13 is an identity, but Measurement 7.14 evaluates it on a declared instance list and at a declared displacement scale; $\log R$ has a pole inside the placement period, so no global statement over the period is available, and none is made. The number 879 is a derivative at a point on named instances, not a rate.

(L11) **What is not claimed.** Problem 7.1 is open; nothing here bounds it and nothing here is conditional on it. No statement of this paper is an arithmetic statement, no comparison against the strength of any conjecture appears anywhere, and Measurement 7.9 claims no limit. Remark 7.15 states this in full, and Table 1 is the binding record of what is claimed at what strength: where the prose and the table disagree, the table governs.

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